

TALKING ABACWS

STAKEHOLDER QUESTION CATALOGUE

Researchers and Data Scientists

STAKEHOLDER 34

80 decision-facing questions across eight evidence domains for defensible building research, data science and responsible communication.

Each card defines what could be measured, linked and analysed - and where evidence, permission, uncertainty or professional authority requires the service to stop.

01 Research questions, hypotheses and study boundaries

02 Data catalogue, variables, coverage and access

03 Sensor placement, calibration and data quality

04 Linkage, timestamps, provenance and versioning

05 Methods, baselines, confounding, bias and uncertainty

06 Privacy, ethics, consent and disclosure control

07 Reproducibility, code, metadata and secure sharing

08 Interpretation, visualisation, publication and impact

Purpose, scope and non-negotiable boundaries

This catalogue is a research-design and evidence-boundary tool. It helps researchers and data scientists formulate answerable questions, identify proposed measurements and authoritative sources, choose defensible methods and state what the service cannot conclude. Every card is a question to investigate, not a claim that a stream, integration, model or dataset currently exists.

PROPOSED UNTIL VERIFIED

Sensors, integrations, models and datasets remain proposed or candidate until deployment, asset mapping, commissioning, permission and current health are independently confirmed.

HISTORICAL PLAN ONLY

The May 2021 floor plan supports historical spatial orientation only. It cannot establish current room use, access, services, hazards or closures.

CURRENT OWNERS PREVAIL

Current authoritative owners and competent professionals decide permissions, operational meaning, safety, compliance and action.

AGGREGATE EVIDENCE ONLY

Use privacy-preserving aggregates. Do not identify, recognise, track or profile people, infer sensitive traits, or reconstruct individual routines.

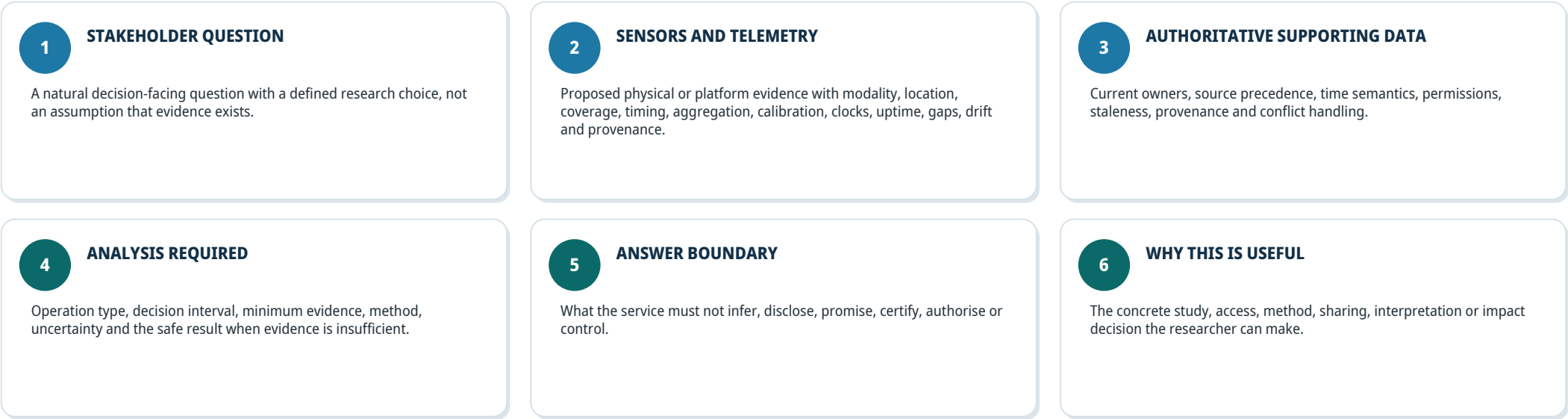
NO CAUSAL SHORTCUTS

Correlation, prediction and anomaly scores do not establish causation, fault, intent, risk, compliance or effectiveness by themselves.

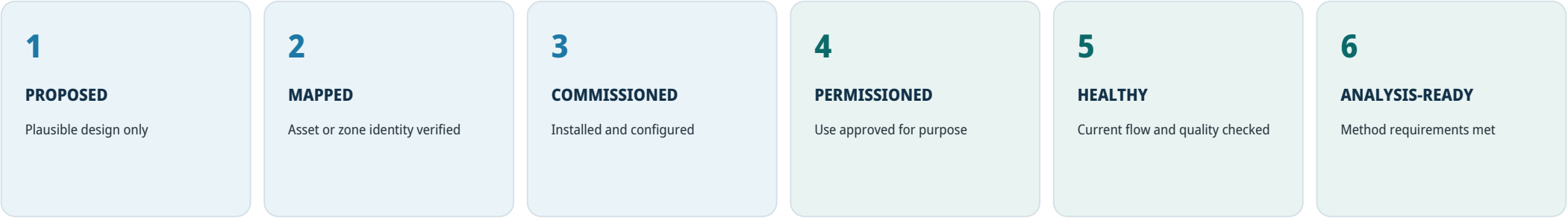
SAFE FAILURE IS VALID

When evidence, permission, mapping or method is inadequate, state what is missing, narrow the claim or decline the analysis.

Field guide and evidence-status model



EVIDENCE STATUS MODEL - EVERY GATE MUST BE EXPLICIT



Catalogue map: 80 questions in eight categories

Each category contains exactly ten question cards. Two cards appear on each catalogue page. The sequence follows the evidence lifecycle from study definition to responsible interpretation and impact.

NO.	CATEGORY	QUESTION IDS	STARTS	PRIMARY DECISION OUTPUT
1	Research questions, hypotheses and study boundaries	RD-001 to RD-010	p. 5	Define a testable estimand, comparison, precision target and transfer boundary.
2	Data catalogue, variables, coverage and access	RD-011 to RD-020	p. 10	Establish which evidence exists, what variables mean, where they cover and who may use them.
3	Sensor placement, calibration and data quality	RD-021 to RD-030	p. 15	Demonstrate representativeness, calibration, timing integrity, continuity and comparability.
4	Linkage, timestamps, provenance and versioning	RD-031 to RD-040	p. 20	Create trustworthy joins across identifiers, time semantics, spatial states and immutable lineage.
5	Methods, baselines, confounding, bias and uncertainty	RD-041 to RD-050	p. 25	Choose methods and baselines that address dependence, confounding, bias and decision uncertainty.
6	Privacy, ethics, consent and disclosure control	RD-051 to RD-060	p. 30	Minimise data, respect approvals and consent, and prevent disclosure or sensitive inference.
7	Reproducibility, code, metadata and secure sharing	RD-061 to RD-070	p. 35	Make extraction, code, environments, tests, sharing and preservation reproducible and secure.
8	Interpretation, visualisation, publication and impact	RD-071 to RD-080	p. 40	Communicate uncertainty, align claims with evidence and plan responsible publication and impact.

Category 1: Research questions, hypotheses and study boundaries

RD-001

STAKEHOLDER QUESTION

CATEGORY 1

For the verified room set, how precisely can aggregate session-use intensity be associated with post-session CO2 recovery after accounting for room, weather and ventilation-state differences?

SENSORS AND TELEMETRY

Proposed evidence: Use research-grade NDIR CO2 with temperature and relative humidity, anonymous bidirectional doorway counts, approved BMS fan or damper states, and an outdoor CO2 and weather reference. Spatial coverage and limits: Select at least six currently verified rooms spanning teaching, research and shared-space conditions; mount room sensors in representative occupied zones, away from supply jets, doors, windows and local emitters, and map every BMS point to its served zone. Temporal coverage and aggregation: Acquire environmental and count data at 1-minute cadence from 30 minutes before until 120 minutes after each eligible session; keep native values and derive predeclared 5-minute medians and recovery episodes. Record-specific quality check: Check CO2 zero and span against a traceable portable reference before and after the campaign, document counter direction error and door-state ambiguity, and require cross-stream offset no greater than 30 seconds. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: facilities and BMS owners; the current space-data owner; timetabling; the sensor-platform steward; the weather-source owner; and the study data steward. Precedence: commissioned BMS-to-zone mappings outrank point names; observed anonymous counts outrank timetable capacity for measured use, while the timetable describes planned sessions only. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: room use and BMS mappings must be confirmed for the study period; health evidence must be no older than seven days at each extraction. Conflict rule: quarantine episodes with unresolved room identity, timetable, count-direction or control-state disagreement and ask the named owner to resolve them.

ANALYSIS REQUIRED

Operation: observation, estimate and comparison; not equipment diagnosis. Decision interval: after a feasibility pilot, at each frozen analytical snapshot and before any confirmatory claim. Minimum evidence: at least six verified rooms, eight representative teaching weeks, 40 complete recovery episodes, at least 80% episode completeness in each retained room and usable outdoor covariates; otherwise justify a precision simulation. Method: fit prespecified nonlinear-decay or segmented mixed-effects models with room and day effects, ventilation-state and weather covariates, lag sensitivity and alternative aggregate-use proxies. Uncertainty: report bootstrap 95% intervals, between-room variance, calibration and clock sensitivity, and bounds under plausible missing episodes. Safe failure: publish descriptive recovery profiles only, narrow the estimand or state that the association is not estimable.

ANSWER BOUNDARY

Do not identify occupants, equate a timetable entry with observed use, convert the decay estimate into a certified ventilation rate, diagnose a building-services fault or describe an association as proof of causation.

WHY THIS IS USEFUL

Lets the team decide whether to proceed to a confirmatory study, extend collection, remap BMS points, replace the aggregate-use proxy or abandon the comparison.

RD-002

STAKEHOLDER QUESTION

CATEGORY 1

Which indoor-environment event definition is stable enough to preregister without turning an operational band into a health, safety or compliance claim?

SENSORS AND TELEMETRY

Proposed evidence: Use only commissioned room-zone measurements relevant to the chosen outcome, such as temperature, relative humidity, CO2, PM2.5, illuminance or A-weighted sound level without audio recording. Spatial coverage and limits: Cover the verified zones named in the protocol and retain device position, height, nearby vents, windows and equipment; do not infer whole-room exposure from a local instrument without a representativeness check. Temporal coverage and aggregation: Retain native samples from 30 seconds to 5 minutes and calculate candidate events with explicit threshold, minimum duration, recovery, hysteresis and aggregation rules. Record-specific quality check: verify units, detection limits, calibration status, warm-up behaviour and threshold-near measurement error; require UTC offset within half the native sample interval. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: the facilities environmental owner; health and safety or another competent professional for any applicable normative interpretation; instrument stewards; and the principal investigator for the research definition. Precedence: current owner-issued operational bands and approved protocol definitions outrank generic defaults; a research threshold never becomes a statutory or clinical threshold by reuse. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: reconfirm bands whenever guidance, room use, averaging period or measurement method changes. Conflict rule: retain competing applicable sources side by side; do not silently choose the one that produces more events.

ANALYSIS REQUIRED

Operation: calculation plus comparison. Decision interval: before outcome lock and whenever a threshold source or sensor method changes. Minimum evidence: verified units and averaging basis, one uncertainty estimate per modality, four representative weeks, at least 30 candidate events or a justified rare-event design, and two or more plausible definitions. Method: compare event frequency, duration, classification agreement and boundary instability across definitions; examine continuous outcomes alongside thresholded versions. Uncertainty: report 95% intervals for event rates, the share of samples whose classification changes under measurement uncertainty, and sensitivity to duration and hysteresis. Safe failure: retain the variable as continuous, label the event definition exploratory, or decline event counting.

ANSWER BOUNDARY

Do not make medical-risk statements, certify statutory performance, select a threshold because it yields significance, or hide uncertainty about applicability, averaging period or sensor accuracy.

WHY THIS IS USEFUL

Creates a preregisterable outcome, prevents threshold shopping and lets later users reconstruct exactly how a continuous stream became an event count.

Category 1: Research questions, hypotheses and study boundaries

RD-003

STAKEHOLDER QUESTION

CATEGORY 1

Which unit of analysis - sensor, zone, room, session, day or shared system - matches the estimand and the genuinely independent replication?

SENSORS AND TELEMETRY

Proposed evidence: Use the proposed study streams plus platform telemetry that identifies sensor, gateway, zone and upstream plant relationships; no additional person sensing is justified. Spatial coverage and limits: Cover every candidate analytical level and the current hierarchy connecting devices to zones, rooms, floors and shared systems; verify co-located devices and common upstream dependencies. Temporal coverage and aggregation: Retain native time series and produce only prespecified summaries at candidate units and intervals; preserve the source count contributing to each aggregate. Record-specific quality check: verify duplicate feeds, shared gateways, common calibration batches, relocation history and UTC alignment; require stable identifiers and explicit gap propagation. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: the sensor-platform steward; facilities asset and BMS owners; the current space-data owner; timetabling where sessions are units; and the study data steward. Precedence: current commissioned asset and zone hierarchies outrank name similarity or geometric proximity; administrative room labels do not prove statistical independence. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: hierarchies must be valid for every analysed period and refreshed after plant, room or sensor changes. Conflict rule: quarantine observations whose parent unit is ambiguous and retain all plausible mappings until the owner decides.

ANALYSIS REQUIRED

Operation: estimate and recommendation. Decision interval: at design lock, after major mapping change, and before final modelling. Minimum evidence: stable identifiers, repeated observations in at least three candidate units at every modelled level, at least 20 independent top-level clusters for cluster-only inference or a justified small-sample method. Method: estimate intraclass correlations, variance components, design effects and effective sample size; compare naive, cluster-robust and multilevel uncertainty. Uncertainty: report mapping ambiguity, confidence intervals for variance components and the sensitivity of conclusions to coarser units. Safe failure: move to a coarser estimand, use descriptive summaries, or state that independent replication is unavailable.

ANSWER BOUNDARY

Do not count correlated sensors or time points as independent samples, infer current boundaries from the historical plan, or derive person-level behaviour from zone or system aggregates.

WHY THIS IS USEFUL

Determines the denominator, prevents falsely narrow intervals and aligns the model with the building structure that actually generated the data.

RD-004

STAKEHOLDER QUESTION

CATEGORY 1

Which observation dates and duration are needed to cover the calendar, weather and operating regimes in scope, and which periods must be excluded?

SENSORS AND TELEMETRY

Proposed evidence: Use the smallest commissioned environmental, occupancy-proxy, energy or BMS stream set needed to recognise the prespecified regimes, together with an approved outdoor weather source. Spatial coverage and limits: Choose verified spaces and systems that represent the target use patterns; include no area solely because the historical plan shows it. Temporal coverage and aggregation: Collect at native cadence, derive daily and weekly coverage summaries, and tag term weeks, weekends, holidays, closures, commissioning periods and weather strata without filling missing regimes synthetically. Record-specific quality check: verify long-run calibration checks, firmware and configuration changes, clock behaviour across daylight-saving changes, seasonal uptime, gaps and monotonic drift. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: timetabling and academic-calendar owners; facilities operations; current space-data owners; weather-source owner; sensor steward; and the principal investigator. Precedence: dated operational and closure records outrank calendar assumptions; owner-confirmed room status outranks the historical plan; observed telemetry defines data presence, not scheduled activity. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: calendar and operational states must cover the full study interval; weather retrievals must be versioned at extraction. Conflict rule: exclude or separately label periods whose operating state, closure status or room use cannot be resolved.

ANALYSIS REQUIRED

Operation: calculation plus recommendation. Decision interval: during pilot completion, monthly during collection, and before stopping data acquisition. Minimum evidence: at least one complete cycle of every named regime, four independent weeks per common stratum, ten usable episodes per rare stratum where feasible, and prespecified precision targets. Method: use stratified coverage tables, variance stabilization curves, simulation-based precision checks and out-of-time validation for any seasonal model. Uncertainty: report precision by stratum, uncertainty for sparsely observed regimes and the effect of excluding abnormal periods. Safe failure: extend collection, narrow the target period, collapse unsupported strata or state that seasonal generalisation is not available.

ANSWER BOUNDARY

Do not promise full-year validity from a short campaign, fill unobserved seasons with model output, or treat closure, commissioning or exceptional-use periods as ordinary operation.

WHY THIS IS USEFUL

Supports a defensible stop-or-continue decision and prevents a convenient data window from being misrepresented as coverage of the intended operating year.

Category 1: Research questions, hypotheses and study boundaries

RD-005

STAKEHOLDER QUESTION

CATEGORY 1

Which current rooms or systems form a defensible comparison set after use, geometry, exposure, services and measurement design are matched?

SENSORS AND TELEMETRY

Proposed evidence: Use proposed environmental and service-state streams only for candidate spaces that have comparable commissioned modalities; add a short temporary co-location campaign where fixed placement differs materially. Spatial coverage and limits: Cover all candidate comparators and document verified volume, orientation, facade exposure, occupancy function, served HVAC zone, sensor position and known local sources. Temporal coverage and aggregation: Use concurrent sampling at matched native cadences and common prespecified aggregates; capture at least two weeks before outcome inspection for design-stage balance checks. Record-specific quality check: verify calibration equivalence or correction, UTC offset within one sample, comparable uptime and gap patterns, and absence of unresolved relocation or firmware changes. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: current space-data and facilities owners; BMS and asset owners; timetabling; sensor stewards; and the study data steward. Precedence: current owner-confirmed function and services outrank room name, nominal capacity or the May 2021 plan; commissioned sensor metadata outrank assumed similarity. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: all structural and use variables must be effective during the comparison interval. Conflict rule: remove candidates with unresolved service-zone, room-use or sensor-method conflicts and document the lost common support.

ANALYSIS REQUIRED

Operation: comparison plus recommendation. Decision interval: before group assignment and again after any material room, service or sensor change. Minimum evidence: complete prespecified balance variables, concurrent coverage, at least four plausible candidates per group where feasible, and common support on major structural covariates. Method: use an outcome-blind design matrix, standardised differences, exact or propensity matching or weighting, and sensitivity to unmeasured confounding. Uncertainty: report residual imbalance, overlap diagnostics, effective sample size and bounds under one or more plausible hidden confounders. Safe failure: provide descriptive contrasts only, choose a different target, or state that no credible comparator exists.

ANSWER BOUNDARY

Do not call spaces equivalent from a shared label, tune comparators after seeing outcomes, conceal poor overlap, or make a causal comparison while major service or measurement differences remain uncontrolled.

WHY THIS IS USEFUL

Lets the team select a defensible comparator or redesign the study before structural imbalance becomes an unfixable source of bias.

RD-006

STAKEHOLDER QUESTION

CATEGORY 1

Does the planned operational change support an interrupted time-series, matched-comparison or difference-in-differences study, or only a descriptive before-and-after account?

SENSORS AND TELEMETRY

Proposed evidence: Use the outcome streams named in the protocol, approved BMS or asset-state telemetry that confirms intervention exposure, and anonymous aggregate use proxies only where necessary. Spatial coverage and limits: Cover treated units and prespecified unaffected comparators served by independently verified systems; verify that sensor positions and modalities remain unchanged across the intervention date. Temporal coverage and aggregation: Acquire continuously for at least the planned pre and post windows, preserve native samples, and calculate fixed daily or session-level outcomes; record exact intervention start, ramp-up and reversal times. Record-specific quality check: perform pre and post calibration checks, require UTC offset under one minute, detect outages, step changes and drift, and retain configuration provenance so measurement change is not mistaken for intervention effect. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: the operational change owner; facilities and BMS owners; current space and timetable owners; sensor steward; and the principal investigator. Precedence: signed or owner-confirmed intervention times and scope outrank planned dates; concurrent work orders, closures and control changes must be retained as competing explanations. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: intervention, comparator and operational records must be effective for the full pre and post interval. Conflict rule: pause causal estimation when exposure timing, concurrent changes or comparator status remains disputed.

ANALYSIS REQUIRED

Operation: estimate with causal interpretation only if identification assumptions hold. Decision interval: at design sign-off, after the pre-period, and after each frozen post-period. Minimum evidence: at least eight independent pre and eight post weeks or a justified higher-frequency design, three treated and three comparison units where feasible, stable measurement, no unresolved concurrent major change, and adequate pre-trend support. Method: compare segmented time series, matched controls and difference-in-differences; test pre-trends, autocorrelation, seasonality, anticipation, ramp-up and placebo dates. Uncertainty: report 95% intervals, model and comparator sensitivity, event-time estimates and bounds for plausible concurrent shocks. Safe failure: report a descriptive before-after comparison, extend the window or state that causal attribution is not supported.

ANSWER BOUNDARY

Do not authorise or control the operational change, claim causation when identification assumptions fail, discard concurrent changes or diagnose safety, compliance or equipment condition from the estimated contrast.

WHY THIS IS USEFUL

Lets the study team choose a causal design, a longer pre-period, a different comparator or an explicitly descriptive evaluation before disruption and analysis cost are committed.

Category 1: Research questions, hypotheses and study boundaries

RD-007

STAKEHOLDER QUESTION

CATEGORY 1

How many independent rooms, days or events are needed to reach the declared precision or decision threshold for this study?

SENSORS AND TELEMETRY

Proposed evidence: Use a limited pilot from the exact proposed modalities and locations, including platform health, gap and calibration telemetry; do not collect additional high-resolution data until the calculation justifies it. Spatial coverage and limits: Sample at least three currently verified units in each important cluster or regime and include the full hierarchy that will define independence. Temporal coverage and aggregation: Retain native pilot samples for two or more representative weeks and calculate the same windows and labels planned for the main study. Record-specific quality check: verify pilot placement, calibration, clock alignment, uptime, gaps, drift and source lineage because optimistic pilot quality will understate the required collection. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: sensor and data-platform stewards; current space and service owners; the principal investigator; and the statistician or method owner. Precedence: pilot estimates from the intended setting outrank generic record counts; justified external priors may supplement but not conceal absent local variance, prevalence or dependence evidence. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: pilot parameters must match the proposed design and be refreshed after outcome, unit, sensor or regime changes. Conflict rule: present competing parameter values and calculate requirements across them rather than selecting a favourable single estimate.

ANALYSIS REQUIRED

Operation: calculation plus recommendation. Decision interval: before protocol lock and after any major change to outcome, unit, model or retention plan. Minimum evidence: pilot estimates of variance, cluster size, autocorrelation, event prevalence, missingness and expected effect or acceptable error, with at least three independent units and two representative weeks. Method: simulate the complete sampling, missingness and modelling pipeline; assess estimation precision, calibration and decision error rather than record count alone. Uncertainty: show requirement ranges across plausible nuisance values and require the target criterion in at least 90% of simulated scenarios. Safe failure: reduce scope, choose a wider precision target, collect a focused pilot or state that feasibility cannot yet be calculated.

ANSWER BOUNDARY

Do not equate time points with independent samples, promise a sample size without assumptions, or use post-hoc power to claim that a null result proves absence of a meaningful effect.

WHY THIS IS USEFUL

Turns a vague request for more data into a defensible collection target and prevents both underpowered analysis and unnecessary long-term retention.

RD-008

STAKEHOLDER QUESTION

CATEGORY 1

Which outcomes, windows and contrasts are primary, secondary or exploratory, and what multiplicity control is defensible before feature generation begins?

SENSORS AND TELEMETRY

Proposed evidence: Use only modalities that have passed the evidence gates for the prespecified outcomes; retain a catalogue of all candidate variables and windows so unselected alternatives remain visible. Spatial coverage and limits: Coverage must match the target units and periods for every claimed outcome; no subgroup may be created from person identification or sensitive-trait inference. Temporal coverage and aggregation: Preserve native samples but compute only locked windows for primary analysis; exploratory windows must be separately labelled and generated after the locked pipeline succeeds. Record-specific quality check: verify common calibration, clock alignment, uptime, gaps, drift and provenance across every outcome so multiplicity control is not applied to incomparable evidence. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: the principal investigator; method lead; data steward; ethics and governance owners where subgroup definitions affect risk; and each source owner. Precedence: the dated protocol and registered analysis specification outrank later convenience; source-owner restrictions outrank a planned analytical contrast. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: the outcome register must be frozen before outcome inspection and versioned after approved amendments. Conflict rule: retain every amendment with reason, date and affected claim; do not overwrite the earlier specification.

ANALYSIS REQUIRED

Operation: calculation plus recommendation. Decision interval: before extraction, before modelling and before each public claim set. Minimum evidence: one explicit primary estimand or a justified small family, named secondary and exploratory sets, expected correlation structure, sample-size implications and a declared family-wise or false-discovery objective. Method: apply hierarchical testing, family-wise correction, false-discovery control or multilevel shrinkage matched to the claim family; run multiverse summaries for exploratory choices. Uncertainty: report adjusted intervals or q-values, the number of attempted analyses and the sensitivity of headline findings to plausible windows. Safe failure: reduce the outcome family, label results exploratory or withhold ranking when multiplicity cannot be controlled.

ANSWER BOUNDARY

Do not select outcomes, windows or subgroups because they are significant, conceal the number of attempted analyses, or infer sensitive participant attributes from aggregate building data.

WHY THIS IS USEFUL

Creates a transparent claim hierarchy, protects confirmatory results from researcher degrees of freedom and still preserves a clearly bounded space for discovery.

Category 1: Research questions, hypotheses and study boundaries

RD-009

STAKEHOLDER QUESTION

CATEGORY 1

Which exclusion, censoring and abnormal-period rules must be locked before outcome inspection, and how will sensitivity to those rules be shown?

SENSORS AND TELEMETRY

Proposed evidence: Use raw streams together with quality flags, platform health, work-order state, approved closure records and sensor configuration history; no new sensing is required solely to justify exclusions. Spatial coverage and limits: Rules must identify the exact device, zone, room, system and interval affected rather than excluding an entire building period by convenience. Temporal coverage and aggregation: Evaluate exclusions at native cadence, then propagate flags to prespecified aggregates with explicit partial-coverage rules; never silently fill excluded intervals. Record-specific quality check: verify flag provenance, calibration and clock alignment, require gap and drift flags to be reproducible, and distinguish sensor failure from a valid extreme observation. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: sensor-platform and facilities owners; work-order and closure owners; current space owner; principal investigator; and data steward. Precedence: owner-confirmed outages, closures and interventions outrank analyst inference; valid extreme measurements remain data unless an objective quality rule excludes them. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: each exclusion source must be current for the affected interval and linked to a dated rule. Conflict rule: unresolved anomaly causes must remain an explicit unknown category rather than being assigned a convenient label.

ANALYSIS REQUIRED

Operation: comparison plus estimate. Decision interval: when the analysis specification is frozen and at every dataset release. Minimum evidence: a named reason, objective trigger, time window and source for every rule, plus retained and removed sample counts and exposure time by unit and regime. Method: apply rules programmatically; compare full, primary and plausible alternative datasets; examine whether removal depends on outcome level, room, time or intervention state. Uncertainty: report exclusion percentages, 95% intervals for key estimates under each set and bias bounds when more than 20% is removed or removal is differential. Safe failure: downgrade to descriptive evidence, collect better event metadata or withhold the affected estimate.

ANSWER BOUNDARY

Do not remove inconvenient observations, invent causes for unexplained anomalies, erase the decision trail, or describe a heavily selected sample as representative of normal operation.

WHY THIS IS USEFUL

Makes data cleaning defensible and shows whether a result survives plausible rules or instead depends on a narrow, selectively retained subset.

RD-010

STAKEHOLDER QUESTION

CATEGORY 1

To which buildings, systems and operating conditions could a result reasonably transfer, and which target settings remain unsupported?

SENSORS AND TELEMETRY

Proposed evidence: Use the study streams plus contextual measurements that define transportability, such as room volume, facade exposure, HVAC type, operating schedule, outdoor conditions and aggregate use intensity; do not add person-level profiling. Spatial coverage and limits: Cover the full Abacws study sample and any approved external comparison sites with harmonised modalities and verified current mappings. Temporal coverage and aggregation: Use matched concurrent windows where possible and fixed daily or episode-level summaries; retain site-specific native data and quality metadata. Record-specific quality check: verify cross-site calibration or co-location, clock and timezone alignment, uptime, gaps, drift, firmware and aggregation equivalence before pooling. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: current building, space, facilities and sensor owners at every site; protocol and data stewards; and any external dataset custodian. Precedence: current site metadata and source-owner definitions outrank assumed equivalence from building labels; Abacws evidence cannot establish unobserved target conditions. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: context variables must be valid for the measurement period and refreshed after refurbishment or service changes. Conflict rule: retain site-specific definitions and analyse unresolved harmonisation differences rather than forcing a common label.

ANALYSIS REQUIRED

Operation: estimate and forecast of transport error. Decision interval: before any external claim and whenever the target population changes. Minimum evidence: an explicit target population, measured effect modifiers, overlap diagnostics, at least two external contexts or a justified mechanistic argument, and enough observations in each supported stratum. Method: use stratified estimates, reweighting, hierarchical partial pooling or transportability models; perform leave-site-out validation when multiple sites exist. Uncertainty: report target-specific intervals, lack-of-overlap regions, between-site heterogeneity and sensitivity to unmeasured effect modifiers. Safe failure: limit the claim to Abacws, present a hypothesis for external testing or state that transfer cannot be estimated.

ANSWER BOUNDARY

Do not generalise from one building because it appears typical, treat users as interchangeable populations, conceal unsupported target strata, or promise performance under unobserved climates, systems or operating practices.

WHY THIS IS USEFUL

Defines where evidence may responsibly travel and helps partners decide whether to adopt, replicate locally or first collect bridging measurements.

Category 2: Data catalogue, variables, coverage and access

RD-011

STAKEHOLDER QUESTION

CATEGORY 2

Which candidate streams can enter the protocol today, which are permissioned but unhealthy, and which remain only catalogue entries or proposals?

SENSORS AND TELEMETRY

Proposed evidence: No new occupant sensing is justified. Use data-catalogue records, API discovery responses, configuration inventories, gateway and pipeline health telemetry, and a small permissioned test extract from each candidate source. Spatial coverage and limits: Limit discovery to approved platforms, named datasets and mapped building assets; do not scan unrelated networks or infer hidden systems. Temporal coverage and aggregation: Capture a status snapshot at request time, seven-day health summaries and the exact test-extract interval; retain native health events and daily completeness aggregates. Record-specific quality check: verify service identity, asset mapping, schema, calibration evidence where physical sensing is involved, UTC clock offset, uptime, gaps, drift flags and immutable source provenance. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: sensor-platform, BMS, metering, space-data, timetable and data-governance owners; system administrators; and the requesting study data steward. Precedence: owner-confirmed commissioning, permission and service status outrank catalogue descriptions, technical plausibility or previous access. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: health checks must be within seven days and permission current for the intended purpose and resolution. Conflict rule: classify unresolved items at the lowest supported gate and do not substitute a similarly named stream.

ANALYSIS REQUIRED

Operation: lookup plus observation. Decision interval: before protocol design, before extraction and at every frozen dataset. Minimum evidence: a six-gate record for proposed, mapped, commissioned, permissioned, healthy and analysis-ready status, plus a successful schema-valid test extract. Method: run automated reachability, row-count, timestamp, schema, range and provenance checks and compare the result with owner declarations. Uncertainty: report gate confidence, expected versus observed coverage and all blocking unknowns. Safe failure: label the stream proposed, unavailable or not analysis-ready and redesign without it.

ANSWER BOUNDARY

Do not imply that a stream exists because it is plausible, bypass access controls, promise live service, or use a proposed signal as if it were measured evidence.

WHY THIS IS USEFUL

Prevents the protocol from depending on imaginary or unusable data and identifies the exact permission, mapping, health, schema or retention gap that must be resolved.

RD-012

STAKEHOLDER QUESTION

CATEGORY 2

What does each variable mean in its current schema, unit, reference frame and missing-value convention, and when did those meanings change?

SENSORS AND TELEMETRY

Proposed evidence: No new sensing is required. Use permissioned raw payload samples, device configuration, schema registry entries and instrument specifications for each proposed variable. Spatial coverage and limits: Cover every device type, source version, room or asset mapping and time interval intended for analysis, including legacy and replacement hardware. Temporal coverage and aggregation: Inspect native cadence and payload precision; calculate no aggregate until units, timestamps, missing codes and state encodings are confirmed. Record-specific quality check: verify calibration units and conversion chains, clock fields, duplicate values, sentinel codes, quantisation, drift flags and device/configuration lineage. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: device and platform stewards; BMS and meter owners; data architects; current space and asset owners; and the study data steward. Precedence: the source-owner schema and commissioned configuration outrank dashboard labels or analyst inference; later schema versions do not retroactively redefine older values. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: definitions must be valid for each observation interval and retrieved from a versioned source. Conflict rule: preserve conflicting definitions, block conversion, and ask the variable owner to issue an effective-dated resolution.

ANALYSIS REQUIRED

Operation: lookup plus calculation. Decision interval: before feature engineering and whenever a source, firmware or schema version changes. Minimum evidence: a field-level dictionary containing physical meaning, unit, scale, reference, precision, valid range, missing convention, state code, source version and effective dates. Method: parse representative payloads, unit-test conversions, compare observed ranges with instrument limits and generate schema-difference tables across versions. Uncertainty: report unresolved semantics, conversion error, quantisation and the proportion of records affected by each version. Safe failure: exclude the field, retain it as opaque status only, or split analysis by incompatible version.

ANSWER BOUNDARY

Do not guess units from values, treat missing codes as zero, mix instantaneous and cumulative quantities, or silently reinterpret historical fields under a newer schema.

WHY THIS IS USEFUL

Gives analysts a defensible data dictionary and prevents unit, encoding and version errors from propagating into features, models and published results.

Category 2: Data catalogue, variables, coverage and access

RD-013

STAKEHOLDER QUESTION

CATEGORY 2

Which current rooms, zones or assets does each stream genuinely cover, and how certain is every source-to-location mapping?

SENSORS AND TELEMETRY

Proposed evidence: Use commissioning labels, asset tags, approved installation records and targeted non-imaging walk-down checks; temporary co-location measurements may test ambiguous fixed-sensor assignments. Spatial coverage and limits: Cover every selected device, gateway, BMS point, meter, room and served zone; record coordinate or room references only at the minimum resolution needed for the study. Temporal coverage and aggregation: Create an effective-dated mapping snapshot before collection and after any move or refurbishment; co-location checks should span at least 24 hours of varied operation. Record-specific quality check: verify physical identity, current location, served area, calibration status, clock alignment, uptime, gaps, drift and relocation provenance. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: facilities asset and BMS owners; current space-data owner; sensor steward; project or estates records owner; and the study data steward. Precedence: current physical verification and commissioned asset records outrank device names, network topology, inferred proximity and the May 2021 plan. The May 2021 floor plan may support historical orientation only; current owner records and competent confirmation govern current use, access, services, hazards and closures. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: mappings must carry valid-from and valid-to times and be confirmed after any room or service change. Conflict rule: mark one-to-many or disputed mappings explicitly and exclude them from room-level inference until the owner decides.

ANALYSIS REQUIRED

Operation: lookup plus observation. Decision interval: before access approval, before room-level analysis and after any installation change. Minimum evidence: a unique effective-dated mapping for every primary device or point, installation evidence, served-zone definition and confidence class. Method: construct a bipartite device-to-space or asset graph, test internal consistency, compare temporary co-location patterns where needed and flag impossible mappings. Uncertainty: report mapping confidence, ambiguous coverage, uninstrumented areas and the share of study exposure affected. Safe failure: limit analysis to verified assets or report building-level evidence without room attribution.

ANSWER BOUNDARY

Do not use the historical floor plan to assert current use, access, services, hazards or closures, infer location from signal strength alone, or disclose sensitive equipment locations unnecessarily.

WHY THIS IS USEFUL

Lets the team state exactly where evidence applies and prevents a mislabeled sensor or service point from creating a false room-level conclusion.

RD-014

STAKEHOLDER QUESTION

CATEGORY 2

What usable time span, native cadence, effective cadence, delivery latency and retention does each stream provide for the intended analysis window?

SENSORS AND TELEMETRY

Proposed evidence: Use source timestamps, gateway arrival logs, storage partitions, retention settings and sample-count telemetry; no additional physical sensing is needed. Spatial coverage and limits: Cover every selected device, source and derived table across the complete requested interval, including known migration and outage periods. Temporal coverage and aggregation: Inspect native inter-arrival times, duplicate timestamps and delivery delay; derive hourly and daily completeness, cadence and latency distributions without assuming nominal settings were achieved. Record-specific quality check: verify UTC conversion, daylight-saving handling, clock resets, sampling configuration, uptime, gaps, drift indicators and ingestion lineage. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: platform and database owners; sensor and gateway stewards; BMS or meter owners; retention-policy owner; and the study data steward. Precedence: observed timestamps and storage evidence outrank nominal cadence or retention documentation; approved retention limits access even when copies appear technically reachable. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: latency and retention evidence must be refreshed at extraction because settings and backfills can change. Conflict rule: separate source absence, delayed arrival, deletion and access denial rather than collapsing them into one missing state.

ANALYSIS REQUIRED

Operation: observation plus calculation. Decision interval: during feasibility, at extraction and before each model fit. Minimum evidence: expected cadence and retention from the owner, observed inter-arrival data for at least seven days per operating regime, complete partition inventory and delay timestamps. Method: calculate completeness, burstiness, duplicate rate, effective sample interval, delivery-latency quantiles and available look-back by source and period. Uncertainty: report confidence for estimated cadence, uncertainty in expected counts and exact unsupported intervals. Safe failure: shorten the study window, aggregate more coarsely, use a batch snapshot or declare the stream unsuitable.

ANSWER BOUNDARY

Do not promise real-time availability from a historical store, infer absence from late delivery, reconstruct deleted records without authority, or conceal periods with no retained evidence.

WHY THIS IS USEFUL

Supports realistic study timing, feature-window selection and storage planning before analysts build methods around cadence or history that the source cannot supply.

Category 2: Data catalogue, variables, coverage and access

RD-015

STAKEHOLDER QUESTION

CATEGORY 2

Which fields are direct observations, commanded values, states, platform summaries, proxies or researcher-derived variables, and how should each be used?

SENSORS AND TELEMETRY

Proposed evidence: Use raw payloads, command and acknowledgement logs, BMS point types, meter registers, platform transformations and derived-feature code; no new sensing is required. Spatial coverage and limits: Cover every field used in an outcome, exposure, covariate, label or quality rule and map it to the originating asset or source table. Temporal coverage and aggregation: Inspect native samples and every aggregation stage; preserve the interval and input rows contributing to summaries and features. Record-specific quality check: verify physical modality, calibration applicability, clock semantics, uptime, gap handling, drift flags and full raw-to-derived provenance. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: instrument and BMS owners; platform engineers; meter and asset owners; model or feature authors; and the study data steward. Precedence: source-owner point type and instrument specification outrank field names; a platform summary cannot be relabelled as a direct measurement. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: classification must be effective-dated and refreshed after schema, firmware or feature-code change. Conflict rule: where owners disagree, retain separate interpretations and block use in scientific claims until resolved.

ANALYSIS REQUIRED

Operation: lookup plus diagnosis of data lineage only. Decision interval: before feature selection and before every release. Minimum evidence: an origin record for each field, including modality, physical quantity or state, transformation formula, input sources, aggregation interval and known proxy limitations. Method: build a typed lineage graph, compare field behaviour with declared type and test transformations on traceable examples. Uncertainty: report classification confidence, transformation error and any proxy-to-target mismatch. Safe failure: exclude or relabel the field and restrict output to the observable quantity.

ANSWER BOUNDARY

Do not treat a command as proof of physical action, a status bit as measured performance, a proxy as its target, or a derived feature as independent evidence.

WHY THIS IS USEFUL

Prevents semantic leakage into models and lets researchers choose variables that support the intended claim rather than merely having convenient names.

RD-016

STAKEHOLDER QUESTION

CATEGORY 2

What is the least-privilege route to obtain only the approved variables, locations, period and resolution required for the study?

SENSORS AND TELEMETRY

Proposed evidence: No new sensing is needed. Use catalogue metadata, access-control telemetry, row and column filters, aggregation services and a small synthetic or non-sensitive test fixture. Spatial coverage and limits: Limit scope to named approved datasets, currently verified locations and the minimum spatial granularity; exclude unrelated rooms, systems and identifiers. Temporal coverage and aggregation: Specify fixed extraction windows, maximum cadence and scheduled refresh; avoid continuous access when a frozen batch answers the question. Record-specific quality check: verify source mapping and health before grant, preserve clock and provenance fields needed for quality, and ensure filtering does not remove required gap or drift indicators. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: data owners; information governance and security teams; platform administrators; ethics approval holder; principal investigator; and data steward. Precedence: owner permission, ethics scope and institutional controls outrank researcher convenience or previous access; denial is authoritative until formally changed. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: permissions must carry purpose, user, fields, resolution, location, interval and expiry and be checked at each extraction. Conflict rule: stop access when policy, owner or technical filters disagree and resolve through the competent owner.

ANALYSIS REQUIRED

Operation: lookup plus recommendation. Decision interval: before access is granted, at renewal and when the research question changes. Minimum evidence: an approved data-minimisation table, threat classification, tested filters, named users, expiry, retention and export route. Method: map each requested field and resolution to a stated analytical need; test positive and negative access cases and compare batch, enclave and code-to-data options. Uncertainty: report residual exposure, denied fields and the reproducibility effect of each restriction. Safe failure: provide metadata only, use a more restricted environment, aggregate further or decline access.

ANSWER BOUNDARY

Do not bypass permissions, broaden access for future convenience, expose raw identifiers when aggregates suffice, or treat technical reachability as authorisation.

WHY THIS IS USEFUL

Reduces privacy and security risk while giving the team a clear, supportable route to the exact evidence needed for the approved analysis.

Category 2: Data catalogue, variables, coverage and access

RD-017

STAKEHOLDER QUESTION

CATEGORY 2

At extraction time, is each selected stream sufficiently current, complete and trustworthy for the declared analysis rather than merely reachable?

SENSORS AND TELEMETRY

Proposed evidence: Use automated source-health, sample-count, calibration-status, clock-offset, gateway, ingestion and schema telemetry plus a signed or hashed test extract. Spatial coverage and limits: Cover every device, asset, table and partition contributing to the planned run; health from a neighbouring device or earlier campaign is not sufficient. Temporal coverage and aggregation: Evaluate the exact analysis interval and a recent seven-day operational window; calculate completeness at native cadence and at the intended aggregate. Record-specific quality check: verify mapping and commissioning, calibration validity, UTC alignment, expected cadence, uptime, duplicates, gaps, stuck values, range errors, drift and source lineage. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: source-system and sensor owners; database or platform administrators; facilities and BMS owners; and the study data steward. Precedence: observed current health outranks catalogue status; owner-declared maintenance and decommissioning states outrank apparent data flow. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: health evidence must be generated at run time or within the protocol-defined maximum age, normally 24 hours for live and seven days for batch sources. Conflict rule: block the run when owner status and automated evidence disagree on a critical gate.

ANALYSIS REQUIRED

Operation: observation plus diagnosis of data quality. Decision interval: immediately before every analytic run and after any source restart or backfill. Minimum evidence: a complete expected-source manifest, current health checks, sample expectations, calibration and mapping status, schema tests and provenance hash. Method: apply fail-fast validation for missing sources, stale partitions, schema drift, clock error, duplicates, impossible ranges, stuck signals and unapproved backfills. Uncertainty: report pass, warning and fail counts, coverage confidence and the exact effect of warnings on planned estimates. Safe failure: stop the pipeline, isolate unaffected sources or produce a clearly labelled quality-only output.

ANSWER BOUNDARY

Do not run because the source worked previously, silently accept stale data, fabricate missing partitions, or downgrade a critical failure merely to complete the analysis.

WHY THIS IS USEFUL

Prevents invalid data from entering a model and creates a reproducible go or no-go decision for each analytic snapshot.

RD-018

STAKEHOLDER QUESTION

CATEGORY 2

What temporal and spatial resolution retains the target signal while minimising unnecessary detail, linkage risk and storage?

SENSORS AND TELEMETRY

Proposed evidence: Use a short, permissioned high-resolution pilot from the exact modality plus lower-resolution candidate aggregates; no extra identifiers or person-level detail are justified. Spatial coverage and limits: Cover representative verified rooms, assets and operating regimes, including at least one high-variability and one low-variability setting. Temporal coverage and aggregation: Retain native pilot samples only inside the approved secure environment and compare fixed temporal windows and spatial groupings; delete or restrict the finer pilot according to permission. Record-specific quality check: verify calibration, clock alignment, uptime, gaps, drift and provenance before attributing differences to aggregation rather than measurement quality. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: source owners; information governance; storage and platform owners; principal investigator; and data steward. Precedence: the scientific need and approved minimisation rule outrank technical maximum resolution; source-owner aggregation semantics must be used where available. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: resolution decisions must be renewed when the outcome, linkage plan, risk class or storage policy changes. Conflict rule: choose the safer coarser option when scientific utility and disclosure risk cannot both be estimated.

ANALYSIS REQUIRED

Operation: comparison plus recommendation. Decision interval: during pilot design and before long-term extraction or retention. Minimum evidence: a named target feature or decision, candidate resolutions, utility metric, storage estimate, disclosure-risk metric and at least two representative weeks. Method: compare signal attenuation, event timing error, model performance, storage and linkage risk across resolutions; use blocked out-of-time tests. Uncertainty: report utility intervals, worst-case rare-event loss, storage range and residual disclosure risk. Safe failure: select the coarsest defensible resolution, narrow the question or keep fine data only inside a controlled environment.

ANSWER BOUNDARY

Do not collect maximum detail by default, claim that aggregation guarantees anonymity, or publish fine-grained room-time traces that enable reconstruction of routines.

WHY THIS IS USEFUL

Balances analytical value, privacy and cost before high-resolution retention becomes entrenched and difficult to justify.

Category 2: Data catalogue, variables, coverage and access

RD-019

STAKEHOLDER QUESTION

CATEGORY 2

Which independent reference measurements or labels can support model training and evaluation, and what error do those references contain?

SENSORS AND TELEMETRY

Proposed evidence: Use traceable portable instruments, approved manual asset-state observations, maintenance confirmations or owner-confirmed operational labels; avoid images, audio and person identification unless separately approved and indispensable. Spatial coverage and limits: Collect references at a stratified sample of the exact rooms, assets and regimes represented in the model, with co-location or explicit spatial offset recorded. Temporal coverage and aggregation: Synchronise reference and candidate measurements at native cadence, define a matching tolerance and retain replicate observations and assessor uncertainty. Record-specific quality check: verify reference calibration and detection limits, UTC alignment, candidate uptime, gaps, drift, label provenance and assessor agreement. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: instrument and asset owners; facilities or competent technical professionals; study label steward; ethics and governance owners where human observation is involved; and principal investigator. Precedence: traceable independent measurement or owner-confirmed state outranks model output or analyst assumption; operational labels remain limited to the authority that issued them. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: reference calibration and label definitions must be valid at observation time. Conflict rule: retain uncertain and conflicting labels rather than forcing consensus; exclude them from gold-standard evaluation or model them probabilistically.

ANALYSIS REQUIRED

Operation: comparison plus estimate. Decision interval: during pilot labelling, before training and before final evaluation. Minimum evidence: prespecified reference method, blinded or independent collection where feasible, at least 100 matched observations or a justified rare-event design, coverage of key regimes and an untouched test subset. Method: estimate measurement error, agreement, sensitivity, specificity or continuous error as appropriate; use repeated labels and hierarchical error models when references are imperfect. Uncertainty: report 95% intervals, inter-assessor agreement, prevalence sensitivity and uncertainty propagated into model performance. Safe failure: treat labels as weak, use unsupervised or descriptive methods, collect better references or decline performance claims.

ANSWER BOUNDARY

Do not call operational guesses ground truth, train and test on the same events, infer a fault without competent confirmation, or hide reference uncertainty behind a single label.

WHY THIS IS USEFUL

Shows whether supervised learning is defensible and identifies where limited labelling effort will most improve trustworthy evaluation.

RD-020

STAKEHOLDER QUESTION

CATEGORY 2

Which single additional measurement, mapping or reference task would reduce the most decision-relevant uncertainty before full collection?

SENSORS AND TELEMETRY

Proposed evidence: Consider only candidate additions tied to a named uncertainty: temporary co-location, an outdoor reference, a current asset mapping, calibration check, anonymous count aggregate or owner-confirmed event label. Spatial coverage and limits: Place additions only in the rooms, assets or regimes where current evidence cannot distinguish competing explanations; avoid building-wide deployment by default. Temporal coverage and aggregation: Run a short pilot long enough to capture the relevant variability, normally one to four weeks, at a cadence matched to the target phenomenon; define the planned aggregate before installation. Record-specific quality check: specify calibration, clock tolerance, expected uptime, gap rule, drift check and full device and commissioning provenance in the acquisition plan. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: principal investigator; method lead; sensor and platform owners; current space and facilities owners; information governance; and budget holder. Precedence: the current decision need, owner permissions and evidence gaps outrank technical novelty; competent owners decide installation and safety feasibility. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: candidate costs, permissions, room status and source health must be current at the decision date. Conflict rule: discard candidates with unresolved ownership, installation or permission conflicts and show the resulting residual uncertainty.

ANALYSIS REQUIRED

Operation: recommendation using calculation and expected value of information. Decision interval: at study design, after pilot results and before any deployment expansion. Minimum evidence: a finite candidate list, current uncertainty model, expected effect on the decision, cost, operational burden, privacy risk and failure probability. Method: simulate posterior or decision-uncertainty reduction, compare expected value of information per unit cost and test whether the added signal changes the chosen action. Uncertainty: report ranges for utility, cost, uptime and model assumptions and rank options only when dominance is stable. Safe failure: recommend no additional sensing, collect a cheaper mapping or calibration record, or state that no candidate is decision-relevant.

ANSWER BOUNDARY

Do not recommend a sensor because it is available or novel, install without owner and safety approval, imply guaranteed information gain, or collect personal data to compensate for weak study design.

WHY THIS IS USEFUL

Directs a limited pilot or metadata task to the evidence gap with the highest expected decision value, avoiding broad collection that adds cost or privacy risk without resolving the main uncertainty.

Category 3: Sensor placement, calibration and data quality

RD-021

STAKEHOLDER QUESTION

CATEGORY 3

Does each fixed environmental sensor represent its intended zone, or mainly a local micro-environment created by vents, doors, windows, equipment or solar gain?

SENSORS AND TELEMETRY

Proposed evidence: Use the fixed modality under study plus two or more traceable portable instruments of the same or better class; record door and window state and approved BMS supply condition where available. Spatial coverage and limits: For each candidate room, test the fixed position and temporary points across occupied-zone heights and plausible gradients, avoiding obstruction and direct supply jets unless those are the target. Temporal coverage and aggregation: Co-sample at 1-minute cadence for at least 48 hours and preferably seven representative days; compare native values and fixed 5-minute summaries across operating states. Record-specific quality check: co-locate instruments before and after the traverse, document response time and shielding, require UTC offset under 15 seconds, track uptime and gaps, and separate spatial offset from calibration drift. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: sensor steward; current space and facilities owners; BMS owner; instrument custodian; and study data steward. Precedence: current physical inspection and commissioned location evidence outrank nominal room association or the historical plan; traceable reference readings outrank visual judgement. The May 2021 floor plan may support historical orientation only; current owner records and competent confirmation govern current use, access, services, hazards and closures. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: position, room function and service state must be valid during every traverse. Conflict rule: retain local-source and access constraints as explicit limitations and do not average away a disputed placement.

ANALYSIS REQUIRED

Operation: comparison plus recommendation. Decision interval: at commissioning, after material room or service change, and before room-level inference. Minimum evidence: one fixed and at least two temporary positions per room, 48 hours covering two operating states, pre and post co-location and complete placement metadata. Method: estimate fixed-versus-zone bias, within-room variance and state-specific gradients with mixed models or blocked paired differences. Uncertainty: report 95% intervals, spatial representativeness error, calibration contribution and the proportion of time conclusions differ by position. Safe failure: limit claims to the local position, relocate or add sensors, or exclude the room from zone-level analysis.

ANSWER BOUNDARY

Do not treat one wall-mounted value as occupant exposure, infer ventilation adequacy, move equipment without owner approval, or hide gradients that challenge the room average.

WHY THIS IS USEFUL

Determines whether a fixed device supports room-level research and identifies the least disruptive placement or sampling change needed.

RD-022

STAKEHOLDER QUESTION

CATEGORY 3

How many sensors and which positions are needed to resolve the spatial gradient relevant to the hypothesis without unnecessary deployment?

SENSORS AND TELEMETRY

Proposed evidence: Use a temporary grid of calibrated portable sensors for the target modality, with the existing fixed device included as an anchor; no imaging or person tracking is needed. Spatial coverage and limits: Select a small stratified set of verified rooms by volume, facade, use and HVAC type; place points at justified horizontal and vertical locations while preserving safe access and airflow. Temporal coverage and aggregation: Sample synchronously at 30-second to 2-minute cadence for at least two representative days per room and derive fixed 5-minute summaries and gradient metrics. Record-specific quality check: perform pre and post co-location, verify clock offset under 10 seconds, require balanced uptime, flag moved or obstructed units, quantify gaps and drift, and retain exact placement and configuration provenance. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: instrument and sensor stewards; facilities and current space owners; health and safety for installation constraints; BMS owner; and principal investigator. Precedence: current safe-placement approval and room configuration outrank geometric convenience or the May 2021 plan; commissioned fixed locations are anchors, not proof of sufficiency. The May 2021 floor plan may support historical orientation only; current owner records and competent confirmation govern current use, access, services, hazards and closures. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: room layout and operating state must be confirmed for each campaign. Conflict rule: remove unsafe or inaccessible positions and recalculate design precision rather than substituting unrecorded locations.

ANALYSIS REQUIRED

Operation: calculation plus recommendation. Decision interval: during placement design and after the temporary campaign. Minimum evidence: at least five temporary positions in large or complex rooms or three in smaller rooms, two operating states, matched samples and a prespecified maximum acceptable spatial error. Method: fit variograms, spatial mixed models or design-based interpolation; run sensor-subset simulations to estimate error as positions are removed. Uncertainty: report cross-validation error, confidence bands for gradients, sensitivity to placement and the probability of exceeding the target error. Safe failure: retain local-only claims, increase temporary sampling or conclude that fixed room-wide estimation is not feasible.

ANSWER BOUNDARY

Do not maximise sensor count without decision value, obstruct routes or services, interpolate across unobserved partitions, or claim precise spatial fields from sparse points.

WHY THIS IS USEFUL

Supports a costed placement plan that achieves a stated spatial precision instead of relying on arbitrary density.

Category 3: Sensor placement, calibration and data quality

RD-023

STAKEHOLDER QUESTION

CATEGORY 3

Is the calibration chain traceable and accurate enough for the smallest difference or threshold the study intends to interpret?

SENSORS AND TELEMETRY

Proposed evidence: Use the proposed fixed sensors, a traceable reference instrument appropriate to each modality and documented zero, span or multi-point checks; for low-cost particulate or gas sensors, include relevant environmental co-variables. Spatial coverage and limits: Test every primary device or a justified stratified batch, including replacements and devices from different purchase or calibration lots, at a controlled site and selected field locations. Temporal coverage and aggregation: Record raw readings at native cadence through stabilisation and at multiple reference levels; repeat before and after the study and at protocol-defined intervals. Record-specific quality check: verify reference certificate and range, warm-up, response time, clock alignment, repeated-measure uptime, gaps, hysteresis, drift and full calibration coefficient provenance. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: instrument custodian; sensor manufacturer or accredited laboratory where applicable; sensor-platform steward; principal investigator; and data steward. Precedence: current traceable reference evidence and approved coefficients outrank factory defaults, dashboard adjustments or undocumented prior corrections. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: certificates and coefficients must cover the observation period and intended range. Conflict rule: block quantitative comparison when coefficients, units or certificate validity conflict and retain raw values unchanged.

ANALYSIS REQUIRED

Operation: calculation plus estimate. Decision interval: before commissioning, at scheduled checks, after suspected drift and before final analysis. Minimum evidence: reference values spanning the study range, at least three levels and repeated readings, pre and post checks, environmental covariates where known and coefficient versioning. Method: fit an error model matched to the sensor physics, assess linearity, hysteresis and response time, and propagate calibration uncertainty into downstream estimates. Uncertainty: report bias, repeatability, expanded uncertainty or 95% intervals, range limits and between-device variation. Safe failure: use the sensor for ordinal or descriptive patterns only, narrow the detectable difference or exclude it.

ANSWER BOUNDARY

Do not call a factory specification a current calibration, correct data without retaining raw values and coefficient versions, or certify measurement performance beyond the reference range.

WHY THIS IS USEFUL

Shows whether the instrument can resolve the planned effect and prevents apparent scientific differences from being calibration artefacts.

RD-024

STAKEHOLDER QUESTION

CATEGORY 3

Is drift materially changing the signal, when did it begin, and should the affected values be corrected, segmented or excluded?

SENSORS AND TELEMETRY

Proposed evidence: Use repeated co-location with a stable reference, redundant neighbouring sensors where physically comparable, internal diagnostics and approved environmental covariates; redundancy alone is not a reference. Spatial coverage and limits: Cover every primary device over its installed range, with checks at commissioning, scheduled intervals and after cleaning, firmware change, relocation or extreme exposure. Temporal coverage and aggregation: Retain native values, calibration checks and daily stability summaries; define a drift alarm in physical units and a minimum sustained duration before looking at outcomes. Record-specific quality check: verify reference stability, UTC alignment, common operating states, uptime and gaps, distinguish abrupt configuration change from gradual drift, and preserve raw and corrected-series provenance. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: sensor and instrument stewards; maintenance owner; platform administrator; principal investigator; and data steward. Precedence: traceable reference checks and owner-confirmed changes outrank agreement with neighbouring sensors; correction coefficients apply only to their effective intervals. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: drift evidence must bracket the analysed period or be explicitly extrapolated with uncertainty. Conflict rule: retain multiple plausible change points when evidence is ambiguous and block a single correction from being treated as certain.

ANALYSIS REQUIRED

Operation: diagnosis of measurement quality plus estimate. Decision interval: monthly, after quality alarms and before each frozen dataset. Minimum evidence: at least three reference checks over time or a justified redundant design, stable comparison conditions, a prespecified materiality threshold and enough pre and post samples. Method: use control charts, change-point detection and hierarchical calibration models; validate any correction on held-out reference periods. Uncertainty: report change-point intervals, correction uncertainty, residual bias and downstream sensitivity with uncorrected, segmented and corrected series. Safe failure: segment the record, exclude the affected interval, use descriptive rank information or state that drift timing is unresolved.

ANSWER BOUNDARY

Do not apply an undocumented correction, erase inconvenient drift, treat agreement with a nearby device as calibration or claim that corrected values recover the unobserved truth exactly.

WHY THIS IS USEFUL

Supports a transparent choice between correction, segmentation, recollection and exclusion before drift distorts trends.

Category 3: Sensor placement, calibration and data quality

RD-025

STAKEHOLDER QUESTION

CATEGORY 3

Can devices measuring the same modality be pooled, or do model, batch, firmware, enclosure or installation effects require separate treatment?

SENSORS AND TELEMETRY

Proposed evidence: Use all candidate devices in a common co-location against a traceable reference, including each model, purchase batch, firmware version and enclosure design represented in the study. Spatial coverage and limits: Run controlled co-location before deployment and field co-location in at least one representative environment; document installation orientation and airflow exposure. Temporal coverage and aggregation: Sample at native cadence through at least three relevant levels or states and for 48 hours in the field; retain device-level raw values and fixed aggregates.

Record-specific quality check: verify identical clocks, reference calibration, warm-up, uptime, gaps, drift, configuration and complete device genealogy. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: sensor and instrument stewards; manufacturer documentation owner; procurement or asset owner; platform administrator; and study data steward. Precedence: observed co-location evidence outranks nominal model specification; owner-confirmed firmware and hardware history outrank device-name parsing. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: comparability evidence must cover every device version and the observation period. Conflict rule: keep incompatible strata separate and do not transfer a correction across batches without evidence.

ANALYSIS REQUIRED

Operation: comparison plus estimate. Decision interval: before pooling, after firmware or hardware change, and before final modelling. Minimum evidence: at least five devices per major stratum where feasible, a traceable reference, three levels or states, 48 field hours and prespecified acceptable bias and variance. Method: estimate device, batch, firmware and environment effects with mixed models; assess equivalence, not only absence of significant difference. Uncertainty: report equivalence intervals, variance components, correction uncertainty and sensitivity of pooled results to device-stratified models. Safe failure: analyse devices separately, include device effects, calibrate by stratum or exclude unsupported versions.

ANSWER BOUNDARY

Do not pool because devices share a label, assume firmware changes are neutral, or use significance failure as proof of equivalence.

WHY THIS IS USEFUL

Determines whether pooling increases evidence or merely hides systematic device differences.

RD-026

STAKEHOLDER QUESTION

CATEGORY 3

Are device and platform clocks aligned closely enough for the joins, lags and event ordering used by the study?

SENSORS AND TELEMETRY

Proposed evidence: Use device clocks, gateway receipt times, NTP or PTP status, BMS historian timestamps and controlled synchronisation events visible across selected streams; no person-related event is needed. Spatial coverage and limits:

Cover every source participating in a join, including gateways, databases and external weather or timetable systems, with timezone and daylight-saving settings recorded. Temporal coverage and aggregation: Measure offset and jitter continuously where supported and at least daily otherwise; retain raw timestamps and derive corrected UTC time only with versioned rules. Record-specific quality check: verify clock source, reset history, monotonicity, acquisition and network latency, uptime, gaps, drift rate and correction provenance. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: platform and network time owners; BMS and source-system administrators; sensor steward; external source owners; and data steward. Precedence: measured synchronisation events and source-owner time semantics outrank timestamp appearance; observation time must not be replaced by ingestion time without explicit justification. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: time configuration and offset evidence must be effective for each interval and refreshed after restart or daylight-saving transition. Conflict rule: quarantine joins whose plausible offset changes event order or exceeds the protocol tolerance.

ANALYSIS REQUIRED

Operation: calculation plus diagnosis of timing quality. Decision interval: at commissioning, daily during collection and before every cross-stream join. Minimum evidence: one common event or trusted time source per stream, offset and jitter estimates, known latency path and a tolerance derived from the shortest causal or lag window. Method: estimate offset, jitter and drift; apply versioned corrections; run join-sensitivity analysis across plausible time shifts. Uncertainty: report offset intervals, unmatched-event rates and the share of estimates altered by timing uncertainty. Safe failure: widen the join window, aggregate more coarsely, restrict to aligned intervals or decline lag and event-order claims.

ANSWER BOUNDARY

Do not reorder events to fit a hypothesis, assume platform timestamps equal observation time or claim lag direction when the combined clock uncertainty is of the same order as the estimated lag.

WHY THIS IS USEFUL

Prevents false leads, lags and intervention timing created by clock error rather than building behaviour.

Category 3: Sensor placement, calibration and data quality

RD-027

STAKEHOLDER QUESTION

CATEGORY 3

What produced each data gap, and which missing intervals may be retained, imputed under stated assumptions, or only reported?

SENSORS AND TELEMETRY

Proposed evidence: Use source and gateway heartbeat telemetry, power and network status, storage and ingestion events, sensor diagnostics and the raw stream itself; no new occupant data is required. Spatial coverage and limits: Cover every affected device, gateway, upstream service and storage partition, with shared dependencies mapped so correlated outages are visible. Temporal coverage and aggregation: Analyse gaps at native cadence, classify their start and end, and derive completeness by hour, day, room and operating regime; retain late arrivals separately from true absence. Record-specific quality check: verify expected cadence, UTC alignment, source uptime, duplicate and buffer behaviour, drift before and after gaps, and provenance of backfills or manual restores. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: sensor-platform, network, power, database and BMS owners; maintenance owner; and study data steward. Precedence: owner-confirmed outage and backfill records outrank analyst inference; absence of an incident entry does not prove the sensor observed normally. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: dependency and incident evidence must cover the exact gap interval and retrieval date. Conflict rule: use an unresolved category when causes conflict and prevent that label from being treated as random missingness.

ANALYSIS REQUIRED

Operation: diagnosis of data quality plus comparison. Decision interval: weekly during collection and before any missing-data treatment. Minimum evidence: expected sample schedule, raw and ingestion timestamps, dependency health, at least 30 observations before and after each material gap, and a cause confidence scheme. Method: classify gaps by source, dependency and regime; test whether missingness relates to measured outcomes or covariates; compare complete-case and plausible imputation scenarios. Uncertainty: report gap duration, cause confidence, differential missingness and sensitivity bounds rather than a single filled series. Safe failure: leave intervals missing, restrict the estimand, collect replacement evidence or state that imputation assumptions are unsupported.

ANSWER BOUNDARY

Do not fabricate values, call missingness random without evidence, merge late data with real-time availability claims, or use a guessed outage cause as an operational diagnosis.

WHY THIS IS USEFUL

Separates technical absence from selective missingness and guides whether to impute, narrow the study or improve system telemetry.

RD-028

STAKEHOLDER QUESTION

CATEGORY 3

Are detection limits, saturation, quantisation, response time and noise compatible with the effects and events the study is designed to observe?

SENSORS AND TELEMETRY

Proposed evidence: Use instrument specifications plus traceable bench or field tests that span the expected range, include step changes and stable plateaus, and retain native unrounded output. Spatial coverage and limits: Test each model and configuration used in the study, including representative temperature, humidity, light, airflow or acoustic conditions that may affect response. Temporal coverage and aggregation: Sample faster than the expected sensor response during tests, then evaluate the planned study cadence and aggregation; include repeated low, mid and high levels. Record-specific quality check: verify reference range, calibration, clock alignment, stable power, uptime, gaps, drift, clipping flags, digital precision and complete test provenance. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: instrument and sensor stewards; manufacturer specification custodian; principal investigator; and data steward. Precedence: observed verified performance in the intended range outranks headline manufacturer resolution; software display precision does not prove physical sensitivity. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: performance evidence must match firmware, configuration and environmental conditions used during the study. Conflict rule: retain conflicting bench and field results and adopt the more conservative usable range until resolved.

ANALYSIS REQUIRED

Operation: calculation plus comparison. Decision interval: before outcome selection and after any sensor or configuration change. Minimum evidence: a traceable reference, repeated points below and above the target effect, step-response tests, at least three devices per type where feasible and a prespecified detectable-change target. Method: estimate limit of detection, limit of quantification, saturation, quantisation error, time constant and noise spectrum; simulate attenuation under planned aggregation. Uncertainty: report confidence intervals, usable range, false-event probability and minimum detectable change after all error sources. Safe failure: choose a coarser claim, different modality or longer aggregation, or state that the target effect is below measurement capability.

ANSWER BOUNDARY

Do not equate displayed decimal places with accuracy, extrapolate beyond tested range, interpret saturated values quantitatively, or hide smoothing that removes short events.

WHY THIS IS USEFUL

Confirms whether the proposed measurement can physically resolve the research question before collection and modelling effort is committed.

Category 3: Sensor placement, calibration and data quality

RD-029

STAKEHOLDER QUESTION

CATEGORY 3

Could installation conditions or electromagnetic, thermal, acoustic or airflow interference create systematic error that resembles the target phenomenon?

SENSORS AND TELEMETRY

Proposed evidence: Use the target sensor together with relevant interference measurements or controlled state records, such as temperature and humidity for optical particles, airflow for gas sensors, electrical load for electromagnetic noise or background light for optical counters. Spatial coverage and limits: Test representative installed locations, cable routes, enclosures, power supplies and nearby sources; include a temporary reference positioned to reduce the suspected interference. Temporal coverage and aggregation: Collect across deliberate safe state changes or naturally occurring strata for at least 48 hours; retain native samples and fixed aggregates. Record-specific quality check: verify reference calibration, clock alignment, common uptime, gaps, drift, enclosure state and installation provenance before attributing differences to interference. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: sensor and facilities owners; competent electrical or mechanical professionals where relevant; BMS owner; instrument steward; and data steward. Precedence: verified installation and competent-owner evidence outrank visual suspicion; a correlated service state is a candidate explanation, not proof of interference. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: installation and nearby-source records must be current for each tested interval. Conflict rule: preserve unresolved competing explanations and do not assign a fault category without competent confirmation.

ANALYSIS REQUIRED

Operation: comparison plus diagnosis of measurement quality. Decision interval: at commissioning, after relocation or enclosure change, and when a state-linked anomaly appears. Minimum evidence: at least two interference states, a reference or negative-control location, repeated transitions, stable clocks and enough samples to separate time trend from state. Method: use blocked experiments where permitted, negative controls, regression with interaction terms and before-after co-location; test residual structure. Uncertainty: report effect intervals, false-event rates and sensitivity to alternative interference covariates. Safe failure: relocate or shield the sensor, include a correction only after validation, restrict the affected state or exclude the device.

ANSWER BOUNDARY

Do not create unsafe test conditions, diagnose building equipment, assume correlation identifies the interference source, or correct data using an unvalidated relationship.

WHY THIS IS USEFUL

Prevents installation artefacts from being mistaken for building dynamics and gives a defensible basis for relocation, shielding or exclusion.

RD-030

STAKEHOLDER QUESTION

CATEGORY 3

After a sensor is moved, replaced, reconfigured or reassigned, can the time series remain comparable, and from which effective time?

SENSORS AND TELEMETRY

Proposed evidence: Use effective-dated asset mapping, old and new devices in a planned overlap, traceable co-location, configuration and firmware telemetry, and the raw series on both sides of the change. Spatial coverage and limits: Cover the old and new positions, served zones and any unchanged reference location; verify current room use and service mapping independently. Temporal coverage and aggregation: Collect at least 48 hours and preferably seven days of overlap under varied states; retain native values and calculate the same fixed aggregates before and after. Record-specific quality check: verify both calibrations, UTC alignment, equivalent cadence, uptime, gaps, drift, response time and complete relocation or replacement provenance. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: sensor and asset owners; facilities and current space owners; platform administrator; maintenance or project owner; and data steward. Precedence: signed or owner-confirmed change times and physical mapping outrank first or last observed record; a reused device identifier does not prove continuity. The May 2021 floor plan may support historical orientation only; current owner records and competent confirmation govern current use, access, services, hazards and closures. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: change records and mappings must carry exact effective times and configurations. Conflict rule: split the identity or interval whenever overlap, mapping or configuration evidence conflicts.

ANALYSIS REQUIRED

Operation: comparison plus estimate. Decision interval: before the change where planned, immediately after, and before longitudinal analysis. Minimum evidence: effective change time, old-new overlap, common reference, calibration and configuration records, at least two operating states and prespecified equivalence limits. Method: test level, variance, response and state-dependent equivalence; estimate bridging uncertainty and compare segmented, adjusted and unadjusted trends. Uncertainty: report equivalence intervals, change-point uncertainty and the effect of any bridge on long-term estimates. Safe failure: treat the series as separate eras, restrict to post-change data or state that longitudinal continuity is unavailable.

ANSWER BOUNDARY

Do not splice series because identifiers match, infer room continuity from the historical plan, overwrite the change boundary, or apply an adjustment outside its documented overlap.

WHY THIS IS USEFUL

Preserves honest longitudinal evidence and prevents maintenance or relocation from masquerading as a research finding.

Category 4: Linkage, timestamps, provenance and versioning

RD-031

STAKEHOLDER QUESTION

CATEGORY 4

How can device, gateway, BMS point, meter, room and asset identifiers be linked across time without assuming that names are unique or stable?

SENSORS AND TELEMETRY

Proposed evidence: Use commissioned asset registries, platform identifiers, gateway association records, meter hierarchies and targeted physical verification; no wireless tracking or person identifiers are needed. Spatial coverage and limits: Cover every source identifier used in the study and all effective-dated parent relationships, including reused names, replacements and many-to-many service zones. Temporal coverage and aggregation: Capture mapping snapshots at extraction and all owner-confirmed change events; preserve raw identifiers and create a study surrogate that never overwrites source identity. Record-specific quality check: verify physical mapping, commissioning, clocked effective dates, uptime and gap context, drift flags where measurements are linked, and complete mapping provenance. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: asset, BMS, metering, sensor-platform and current space-data owners; project records owner; and study data steward. Precedence: current commissioned registries and physical verification outrank string similarity, network proximity and the May 2021 plan; source identifiers retain their owner-defined meaning. The May 2021 floor plan may support historical orientation only; current owner records and competent confirmation govern current use, access, services, hazards and closures. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: every relationship must have valid-from and valid-to times spanning the observation interval. Conflict rule: allow unresolved one-to-many mappings in the graph but block them from deterministic joins.

ANALYSIS REQUIRED

Operation: lookup plus calculation. Decision interval: before extraction, after any asset change and before every cross-source join. Minimum evidence: unique source identifiers, effective-dated edges, owner and evidence for each edge, collision tests and a complete mapping for every primary record. Method: build a temporal identifier graph, test uniqueness and referential integrity, and quantify join coverage under strict and permissive mappings. Uncertainty: report mapping confidence, unmatched rates, collisions and the share of results sensitive to ambiguous edges. Safe failure: join at a coarser verified level, exclude ambiguous records or state that linkage is unavailable.

ANSWER BOUNDARY

Do not infer identity from similar names, reuse an old room mapping after a change, expose sensitive asset locations unnecessarily, or force one-to-one links where the physical system is shared.

WHY THIS IS USEFUL

Creates stable, time-aware joins and prevents silent identity collisions from contaminating longitudinal analysis.

RD-032

STAKEHOLDER QUESTION

CATEGORY 4

How should streams with different time zones, daylight-saving rules, clock quality and timestamp precision be aligned for this analysis?

SENSORS AND TELEMETRY

Proposed evidence: Use raw source timestamps, timezone and offset fields, clock-health telemetry, gateway receipt times and shared synchronisation events; retain source-local time as evidence. Spatial coverage and limits: Cover all internal and external sources participating in the intended join, including historical partitions that may use different conventions. Temporal coverage and aggregation: Convert to UTC through versioned rules, preserve original precision, and calculate aligned windows no finer than the least reliable source unless the method models timing error. Record-specific quality check: verify clock source and resets, offset and jitter, delivery latency, uptime, gaps, drift and correction provenance for every interval. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: source-system administrators; network time owner; sensor and BMS stewards; external data custodians; and data steward. Precedence: owner-declared time semantics and measured clock evidence outrank file format or timestamp length; ambiguous local times at daylight-saving transitions remain ambiguous until resolved. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: timezone rules and clock configuration must be versioned for the observation date, not only the retrieval date. Conflict rule: retain both plausible UTC mappings for ambiguous records and exclude them from order-sensitive analysis.

ANALYSIS REQUIRED

Operation: calculation plus comparison. Decision interval: before linkage and whenever timezone, firmware or clock configuration changes. Minimum evidence: original timestamps, timezone rule, precision, measured or bounded offset, latency model and a tolerance tied to the research window. Method: convert with tested timezone libraries, estimate residual offsets from shared events and compare joins under plausible shifts and coarser windows. Uncertainty: report timestamp precision, offset intervals, ambiguous-record counts and sensitivity of lag estimates. Safe failure: aggregate more coarsely, use date-only linkage or decline temporal ordering and lag claims.

ANSWER BOUNDARY

Do not infer timezone from building location alone, silently resolve repeated or skipped local times, or claim event order finer than combined clock uncertainty.

WHY THIS IS USEFUL

Makes cross-source timing defensible and prevents daylight-saving or precision errors from appearing as behavioural lag.

Category 4: Linkage, timestamps, provenance and versioning

RD-033

STAKEHOLDER QUESTION

CATEGORY 4

Which timestamp - observation, event start, state validity, source update, ingestion or retrieval - answers each part of the research question?

SENSORS AND TELEMETRY

Proposed evidence: Use raw payload timestamps, event and state fields, BMS historian metadata, database commit or ingestion times and extraction manifests; no additional sensing is required. Spatial coverage and limits: Cover every source type and derived table, including intervals where one timestamp is missing or created downstream. Temporal coverage and aggregation: Retain all available time fields at native precision; derive event windows only from the time semantics named in the protocol and preserve open-ended state intervals. Record-specific quality check: verify source clocks, time-field origin, latency, uptime, gaps, drift and the transformation lineage that created any derived timestamp. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: source-system owners; BMS and sensor stewards; database administrators; timetable or work-order owners; and data steward. Precedence: source-owner semantics outrank analyst naming; ingestion or retrieval time may describe availability but cannot substitute for physical observation without a stated latency model. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: time definitions and field availability must be effective-dated by schema version. Conflict rule: block the affected question when sources disagree about which event a timestamp represents.

ANALYSIS REQUIRED

Operation: lookup plus recommendation. Decision interval: during schema assessment and before every temporal feature or availability claim. Minimum evidence: a time-semantics dictionary for each field, examples across versions, latency evidence and a mapping from research question to permissible time type. Method: construct temporal state intervals, test ordering constraints and calculate differences among observation, ingestion and retrieval times. Uncertainty: report latency distributions, missing time fields and uncertainty introduced by using a proxy time. Safe failure: restrict the claim to data availability, use interval-censored time or decline event-order analysis.

ANSWER BOUNDARY

Do not replace an unknown observation time with ingestion time, treat state-update time as state start without evidence, or hide late arrival in a real-time claim.

WHY THIS IS USEFUL

Clarifies what the timestamps actually support and prevents a technically valid join from answering the wrong temporal question.

RD-034

STAKEHOLDER QUESTION

CATEGORY 4

How should irregular streams be resampled or aggregated without inventing observations, shifting events or changing the estimand?

SENSORS AND TELEMETRY

Proposed evidence: Use the native raw streams, expected-cadence metadata and quality flags; no additional sensing is needed unless irregularity reflects an unresolved device issue. Spatial coverage and limits: Cover all selected devices and periods, stratified by modality because event-driven states, cumulative meters and continuous sensors require different handling. Temporal coverage and aggregation: Preserve native records; define fixed windows, alignment origin, minimum coverage and statistic per modality; store contributing counts, durations and gap flags with every aggregate. Record-specific quality check: verify units, calibration status, clocks, duplicates, uptime, gaps, drift and raw-to-aggregate provenance before combining streams. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: sensor, BMS, meter and platform owners; schema owner; method lead; and data steward. Precedence: source-owner measurement semantics outrank a generic resampling rule; cumulative registers, commands and event states must not be averaged like continuous observations. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: aggregation definitions must be versioned and valid for the source schema and research question. Conflict rule: keep incompatible modalities or schema eras separate and expose partial windows rather than filling them silently.

ANALYSIS REQUIRED

Operation: calculation plus comparison. Decision interval: before feature generation and after any cadence or schema change. Minimum evidence: native records, expected cadence, modality class, prespecified window and origin, coverage rule and at least two plausible aggregation choices for sensitivity. Method: apply time-weighted summaries, state-duration calculations, counter differencing or event counts as appropriate; compare aliasing and boundary effects. Uncertainty: report contributing observations, coverage, aggregation error and sensitivity of results to window alignment and statistic. Safe failure: retain native event analysis, choose a coarser window, exclude partial periods or state that common aggregation is unsupported.

ANSWER BOUNDARY

Do not interpolate across long gaps, average cumulative totals or categorical states, hide partial-window coverage, or select window boundaries to strengthen a pattern.

WHY THIS IS USEFUL

Produces aggregates with a clear physical meaning and prevents resampling convenience from creating synthetic evidence.

Category 4: Linkage, timestamps, provenance and versioning

RD-035

STAKEHOLDER QUESTION

CATEGORY 4

Can sensor data be linked to timetables, weather, work orders, closures and BMS states without false matches or personal-data leakage?

SENSORS AND TELEMETRY

Proposed evidence: Use commissioned sensor streams, approved anonymous aggregate counts where essential, external weather, BMS state and source-system records; no names, device identities or individual schedules are required. Spatial coverage and limits: Link only currently verified rooms, assets and served zones within the approved study scope; use coarse location where a finer mapping adds risk but no analytical value. Temporal coverage and aggregation: Define interval joins with explicit tolerance and state-overlap rules; preserve native timestamps and join status for every row. Record-specific quality check: verify mapping, calibration relevance, clock alignment, uptime, gaps, drift and provenance across all joined sources. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: timetabling, facilities, work-order, closure, BMS, weather and sensor owners; information governance; and data steward. Precedence: owner-confirmed room, event and service mappings outrank string similarity; observed environmental data and planned timetable data retain different meanings. The May 2021 floor plan may support historical orientation only; current owner records and competent confirmation govern current use, access, services, hazards and closures. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: permissions and mappings must cover the exact purpose, period and granularity of the join. Conflict rule: quarantine many-to-many or contradictory matches and retain unmatched records rather than forcing a link.

ANALYSIS REQUIRED

Operation: lookup plus calculation. Decision interval: at linkage design, at every extraction and after source updates. Minimum evidence: effective-dated identifiers, defined join keys, time tolerance based on clock uncertainty, source-specific semantics, privacy assessment and a labelled sample of true and false matches. Method: use temporal interval joins, deterministic mappings and probabilistic matching only after performance is established on labelled pairs; calculate precision, recall, match multiplicity and unmatched exposure. Uncertainty: report match confidence, linkage error intervals and sensitivity of estimates to strict versus permissive joins. Safe failure: join at a coarser level, analyse sources separately or state that linkage is not reliable enough.

ANSWER BOUNDARY

Do not join on a room name alone, expose personal timetable or work-order content, treat a planned session as observed occupancy or retain ambiguous matches simply to increase sample size.

WHY THIS IS USEFUL

Enables richer context while keeping linkage error and privacy risk visible enough to decide whether the combined analysis is defensible.

RD-036

STAKEHOLDER QUESTION

CATEGORY 4

Can every reported value be traced back to the exact raw records, mappings, coefficients, parameters and code that produced it?

SENSORS AND TELEMETRY

Proposed evidence: No new physical sensing is required. Use immutable raw-object identifiers, checksums, extraction manifests, transformation telemetry, calibration-coefficient versions and code commit identifiers. Spatial coverage and limits: Cover every raw source, intermediate table, feature, model input, prediction, table and figure used in the study. Temporal coverage and aggregation: Capture lineage at each run and each transformation boundary; retain row or partition references proportionate to sensitivity and scale. Record-specific quality check: verify source mapping, clock and quality fields are carried forward, gaps and drift flags are not dropped, and provenance records themselves are complete and tamper-evident. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: data-platform and repository administrators; source owners; code and model owners; calibration steward; principal investigator; and data steward. Precedence: immutable source and run identifiers outrank filenames or analyst memory; owner restrictions determine which lineage details may be shared. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: manifests must be created at run time and preserved with the output version. Conflict rule: fail the output when hashes, input versions or transformation steps conflict or are missing.

ANALYSIS REQUIRED

Operation: lookup plus diagnosis of lineage completeness. Decision interval: for every run and before any result is cited or released. Minimum evidence: hashes for all inputs and outputs, code and environment identifier, parameter set, mapping and calibration versions, execution time and successful lineage traversal for all headline values. Method: construct a directed acyclic lineage graph, verify checksums and replay a sampled set of output-to-input traces. Uncertainty: report lineage coverage, unresolved edges and whether missing provenance can alter interpretation. Safe failure: withhold the result, recreate it from a verified snapshot or label it non-reproducible and non-citable.

ANSWER BOUNDARY

Do not infer lineage from folder names, overwrite raw or intermediate evidence, expose restricted paths or identifiers in public outputs, or cite a value with an incomplete trace.

WHY THIS IS USEFUL

Allows independent verification, targeted correction and trustworthy reuse without requiring readers to rely on undocumented analyst memory.

Category 4: Linkage, timestamps, provenance and versioning

RD-037

STAKEHOLDER QUESTION

CATEGORY 4

How should room boundaries, uses, capacities, service zones and access states be versioned when the May 2021 floor plan is historical evidence only?

SENSORS AND TELEMETRY

Proposed evidence: Use current owner-issued space records, effective-dated asset and BMS mappings, approved non-imaging walk-down confirmation and access or closure records; sensing cannot establish legal or operational status. Spatial coverage and limits: Cover only study locations and their relevant adjacency or served-zone relationships; store no unnecessary security-sensitive detail. Temporal coverage and aggregation: Create a spatial-state snapshot for each analysis interval and record every owner-confirmed change time; never retroactively apply the current state to older observations. Record-specific quality check: verify asset and sensor mapping, commissioning, clocked effective dates, uptime and gaps for linked telemetry, drift flags and spatial-version provenance. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: current space-data owner; facilities and BMS owners; access-control or security owner; timetabling where use is planned; project records owner; and data steward. Precedence: current authoritative records and competent professionals outrank the May 2021 plan, sensor observations and informal labels for room use, access, services, hazards or closures. The May 2021 floor plan may support historical orientation only; current owner records and competent confirmation govern current use, access, services, hazards and closures. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: every spatial attribute must carry valid-from and valid-to times and a current confirmation date. Conflict rule: preserve competing states and block operational interpretation until the responsible owner resolves them.

ANALYSIS REQUIRED

Operation: lookup plus comparison. Decision interval: before spatial aggregation, after any building change and at dataset freeze. Minimum evidence: an effective-dated space identifier, geometry or zone reference, current use, capacity if needed, service and access relationships, owner and confidence for every analysed period. Method: build temporal spatial joins, check topological and hierarchy consistency, and compare results under plausible disputed mappings. Uncertainty: report unsupported periods, mapping confidence and the share of findings affected by spatial-state uncertainty. Safe failure: use a coarser building or floor level, exclude disputed periods or state that current spatial attribution is unavailable.

ANSWER BOUNDARY

Do not use the historical plan to declare current room use, access, services, hazards or closures, let sensors override competent owners, or publish sensitive access details.

WHY THIS IS USEFUL

Prevents historical geometry from being mistaken for current operational truth and makes longitudinal spatial analysis honest.

RD-038

STAKEHOLDER QUESTION

CATEGORY 4

How should schema changes, unit corrections, firmware changes and calibration coefficients be versioned so earlier data remain interpretable?

SENSORS AND TELEMETRY

Proposed evidence: Use raw payloads, schema-registry history, firmware and configuration telemetry, calibration records and representative extracts from every era; retain raw values unchanged. Spatial coverage and limits: Cover every source and field used in analysis, including devices or tables introduced, removed or redefined during the study. Temporal coverage and aggregation: Assign effective times to each version and generate compatible derived views only through explicit, tested transformations; preserve native cadence and precision. Record-specific quality check: verify calibration applicability, clock semantics, uptime, gaps, drift flags and complete version-to-observation provenance. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: source-system and sensor owners; schema and platform administrators; calibration steward; code owner; and data steward. Precedence: effective-dated owner-issued schemas and configurations outrank latest documentation; corrected units do not erase how older values were stored. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: version records must span the full observation interval and be retrieved from controlled sources. Conflict rule: split eras and block harmonisation when transformations or effective dates conflict.

ANALYSIS REQUIRED

Operation: lookup plus calculation. Decision interval: at ingestion, after any change and before combining eras. Minimum evidence: version identifiers, effective dates, field-level differences, conversion formulas, tests, raw examples and coverage counts for every selected era. Method: generate schema diffs, unit and range tests, dual-read comparisons and backward-compatibility checks; propagate coefficient uncertainty. Uncertainty: report affected record counts, conversion error, unresolved incompatibilities and sensitivity to era-specific modelling. Safe failure: analyse versions separately, exclude an era or state that a harmonised series cannot be created.

ANSWER BOUNDARY

Do not overwrite raw historical values, apply current coefficients before their valid date, mix incompatible encodings, or hide a structural break behind a common column name.

WHY THIS IS USEFUL

Keeps long-running datasets interpretable and prevents software or calibration changes from becoming invisible scientific discontinuities.

Category 4: Linkage, timestamps, provenance and versioning

RD-039

STAKEHOLDER QUESTION

CATEGORY 4

How can a citable analytical snapshot remain immutable when late data, corrected mappings or source deletions occur after extraction?

SENSORS AND TELEMETRY

Proposed evidence: Use immutable extraction manifests, source partition hashes, watermarks for late arrival, tombstone or correction records and a permissioned frozen copy or reproducible query. Spatial coverage and limits: Cover all sources and derived tables contributing to the result, including external files and owner-issued mapping corrections. Temporal coverage and aggregation: Record extraction and retrieval times, source watermarks and allowed lateness; create a new snapshot rather than mutating the cited one. Record-specific quality check: verify source mapping, calibration and clock fields, uptime and gap status, drift flags and complete snapshot lineage. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: source and platform owners; repository administrator; principal investigator; records or retention owner; and data steward. Precedence: the frozen manifest defines the cited evidence; later owner corrections create a superseding snapshot but do not silently alter the original result. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: source-retention and deletion obligations remain authoritative even when reproducibility would benefit from preservation. Conflict rule: record deletions, corrections and supersession links explicitly and block claims that depend on inaccessible or unverifiable inputs.

ANALYSIS REQUIRED

Operation: lookup plus comparison. Decision interval: at every extraction, before citation and after material late data or correction. Minimum evidence: complete input manifest, hashes, query and parameters, allowed-lateness rule, correction ledger and a rerun on the superseding snapshot. Method: compare record counts, key summaries and headline estimates between snapshots; classify changes as immaterial, interpretive or claim-changing using prespecified tolerances. Uncertainty: report exact differences and uncertainty in changed estimates, not only a version number. Safe failure: retain the original claim with a dated limitation, issue a corrected result or withdraw the unsupported claim.

ANSWER BOUNDARY

Do not mutate a cited snapshot, ignore owner-mandated deletion, conceal late corrections, or call a rerun identical when inputs or mappings changed materially.

WHY THIS IS USEFUL

Creates an immutable evidence object that readers can cite and reproduce while allowing later corrections to be issued as separately identified snapshots rather than silently changing past results.

RD-040

STAKEHOLDER QUESTION

CATEGORY 4

Is provenance complete and internally consistent enough to trust the dataset when owners, identifiers, timestamps, mappings or versions disagree?

SENSORS AND TELEMETRY

Proposed evidence: Use source manifests, owner identifiers, access decisions, mapping versions, calibration and configuration records, quality flags and cryptographic hashes; no additional physical sensing is required. Spatial coverage and limits: Cover every selected source, record partition and transformation boundary, with special attention to copied files and manually supplied data. Temporal coverage and aggregation: Evaluate provenance at acquisition and each frozen run; preserve the original conflicting records and the time each was observed or retrieved. Record-specific quality check: verify mapping, calibration, clocks, uptime, gaps, drift and lineage metadata as first-class evidence rather than optional annotations. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: each source owner; platform and repository administrators; principal investigator; information governance; and data steward. Precedence: current competent owners decide operational meaning and permission; immutable source evidence outranks analyst recollection; no source may silently override another outside its authority. The May 2021 floor plan may support historical orientation only; current owner records and competent confirmation govern current use, access, services, hazards and closures. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: owner and permission assertions must be current at extraction, while historical assertions remain effective-dated. Conflict rule: route each conflict to the owner of the disputed attribute, retain unresolved alternatives and quarantine affected outputs.

ANALYSIS REQUIRED

Operation: diagnosis of provenance quality plus recommendation. Decision interval: before analysis-ready status and before release. Minimum evidence: named owner, source, effective and observed times, permission, version or hash, mapping, calibration and transformation lineage for every critical field. Method: apply provenance completeness rules, authority matrices, temporal consistency checks and graph traversal; sample evidence against source systems. Uncertainty: report completeness by field and source, conflict counts, authority gaps and the maximum share of outputs affected. Safe failure: downgrade the dataset, exclude conflicted evidence or state that no defensible answer can be produced.

ANSWER BOUNDARY

Do not resolve a conflict by choosing the source that supports the desired result, overwrite contradictory lineage or declare the dataset trustworthy while ownership, identity or time semantics remain unresolved.

WHY THIS IS USEFUL

Creates a transparent trust decision and shows exactly which owner or evidence gap prevents analysis-ready status.

Category 5: Methods, baselines, confounding, bias and uncertainty

RD-041

STAKEHOLDER QUESTION

CATEGORY 5

Which baseline best represents the reference or counterfactual condition, and under which changes does that baseline cease to be credible?

SENSORS AND TELEMETRY

Proposed evidence: Use the same commissioned outcome and covariate modalities in baseline and target periods, together with approved BMS state, weather and aggregate use context needed to identify regime changes. Spatial coverage and limits: Cover the same verified rooms, assets or matched comparison units; record any sensor move, refurbishment, schedule change or service-zone change that can break comparability. Temporal coverage and aggregation: Retain native data and calculate identical prespecified intervals in baseline and target periods; include enough pre-period history to characterise trend and seasonality. Record-specific quality check: verify calibration equivalence, UTC alignment, uptime, gaps, drift, firmware and placement continuity across periods. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: facilities and BMS owners; current space and timetable owners; sensor steward; weather owner; intervention owner where relevant; and principal investigator. Precedence: owner-confirmed operating state and change dates outrank calendar assumptions; a historical average does not remain a baseline after material building or measurement change. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: baseline metadata must be effective for its observation period and refreshed when the target question changes. Conflict rule: split or discard baseline periods with unresolved regime or measurement discontinuity.

ANALYSIS REQUIRED

Operation: comparison plus recommendation. Decision interval: at protocol design, after pilot collection and before each effect estimate. Minimum evidence: a named counterfactual interpretation, at least four independent baseline weeks per common regime or a justified alternative, stable measurement and adequate covariate overlap. Method: compare historical, matched, rolling and model-based baselines using pre-period fit, placebo tests and out-of-time prediction. Uncertainty: report baseline prediction intervals, pre-trend uncertainty, overlap and sensitivity of the result to each credible baseline. Safe failure: use descriptive change only, collect a stronger comparator or state that the counterfactual is not identifiable.

ANSWER BOUNDARY

Do not choose the baseline that gives the largest effect, assume the previous year is comparable, or describe a modelled reference as observed reality.

WHY THIS IS USEFUL

Lets the team select a baseline that matches the decision claim and shows when a seemingly simple before-after contrast is not defensible.

RD-042

STAKEHOLDER QUESTION

CATEGORY 5

What missing-data mechanism is plausible for each stream, and which values may be imputed without creating false evidence?

SENSORS AND TELEMETRY

Proposed evidence: Use raw streams, gateway and ingestion health, work orders, calibration and drift flags, and the covariates available before each missing interval; no extra person data are justified. Spatial coverage and limits: Cover every source and cluster used in the analysis and map shared dependencies so correlated gaps are not treated as independent. Temporal coverage and aggregation: Classify missingness at native cadence, then apply prespecified rules at the analytical interval; retain imputation indicators and never overwrite observed values. Record-specific quality check: verify expected cadence, clocks, uptime, gap boundaries, drift before and after gaps and provenance of late or backfilled records. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: source-platform, network, facilities and maintenance owners; sensor steward; method lead; and data steward. Precedence: owner-confirmed outages and source mechanics outrank statistical convenience; an unexplained gap is not automatically missing at random. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: cause evidence must cover the exact interval and be current at analysis. Conflict rule: retain unresolved causes and evaluate multiple mechanisms rather than assigning a single unsupported category.

ANALYSIS REQUIRED

Operation: estimate and comparison. Decision interval: before imputation, at each dataset freeze and whenever missingness exceeds the protocol threshold. Minimum evidence: missingness maps by unit, time, regime and outcome level; at least 20 observed neighbours for local models where used; auxiliary variables available at prediction time; and explicit maximum gap lengths. Method: compare complete-case, inverse-probability weighting, multiple imputation, state-space or no-imputation strategies matched to the mechanism; use masked-value validation. Uncertainty: report imputation error, between-imputation variance, fraction of information due to missingness and sensitivity bounds under not-at-random scenarios. Safe failure: leave values missing, narrow the estimand or state that the result is too dependent on unverifiable assumptions.

ANSWER BOUNDARY

Do not interpolate long gaps by default, use future values in a forecast setting, hide which values are imputed, or claim that imputation restores information that was never observed.

WHY THIS IS USEFUL

Supports an honest missing-data strategy and reveals whether more robust collection is preferable to increasingly assumption-heavy modelling.

Category 5: Methods, baselines, confounding, bias and uncertainty

RD-043

STAKEHOLDER QUESTION

CATEGORY 5

How should temporal autocorrelation, repeated rooms, shared plant and spatial dependence be modelled so uncertainty is not understated?

SENSORS AND TELEMETRY

Proposed evidence: Use the planned streams plus current device-to-room-to-system hierarchy and sampling metadata; no additional sensing is needed unless the hierarchy is unresolved. Spatial coverage and limits: Cover all units and shared dependencies represented in the claim, including sensors on common gateways or services that may fail together. Temporal coverage and aggregation: Retain native timestamps and predeclared analytical windows; preserve cluster identifiers and the number of raw observations contributing to each summary. Record-specific quality check: verify mapping, calibration comparability, clock alignment, uptime, gaps, drift and dependency provenance before estimating dependence. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: sensor-platform, BMS, asset and space owners; method lead; principal investigator; and data steward. Precedence: current commissioned hierarchies outrank room labels and nominal independence; repeated time points cannot create new top-level units. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: hierarchies and sampling settings must be valid for every analysed period. Conflict rule: retain multiple plausible clustering structures and use the most conservative supported one when unresolved.

ANALYSIS REQUIRED

Operation: estimate and comparison. Decision interval: at model design and before each inferential result. Minimum evidence: at least 20 top-level clusters for simple cluster-robust inference or a justified small-sample correction, repeated observations across key regimes and enough series length to estimate autocorrelation. Method: use multilevel, generalized estimating equation, state-space or block-bootstrap methods; compare residual autocorrelation and variance under alternative cluster structures. Uncertainty: report effective sample size, intraclass correlations, autocorrelation intervals and sensitivity of conclusions to cluster choice. Safe failure: aggregate to a coarser unit, use descriptive intervals or state that inferential precision cannot be estimated reliably.

ANSWER BOUNDARY

Do not treat minute samples as independent, ignore shared plant or gateway dependence, or choose a covariance structure only because it narrows intervals.

WHY THIS IS USEFUL

Produces realistic uncertainty and prevents apparent precision from being driven by repeated measurements rather than independent evidence.

RD-044

STAKEHOLDER QUESTION

CATEGORY 5

Which weather, calendar, room, service and aggregate-use factors could confound the association, and are they measured well enough to adjust for?

SENSORS AND TELEMETRY

Proposed evidence: Use only plausible common-cause evidence: approved outdoor weather, room and service metadata, BMS operating states and anonymous aggregate use proxies; avoid person profiling and variables affected by the outcome. Spatial coverage and limits: Cover the same verified units and intervals as exposure and outcome, including shared upstream systems and relevant external conditions. Temporal coverage and aggregation: Acquire or retrieve covariates at source cadence and align them to prespecified windows without using future information; preserve missing and stale states. Record-specific quality check: verify modality, mapping, calibration where physical, clock alignment, uptime, gaps, drift and provenance for every proposed confounder. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: facilities, BMS, weather, timetable, current space and sensor owners; principal investigator; subject-matter expert; and data steward. Precedence: a causal rationale and owner-confirmed meaning outrank predictive association; current operational records outrank historical assumptions. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: confounder definitions and mappings must be valid for the full study interval. Conflict rule: retain disputed variables as sensitivity factors and do not adjust for them as if their role were established.

ANALYSIS REQUIRED

Operation: estimate and recommendation. Decision interval: during causal diagram development, before model fitting and when new competing explanations emerge. Minimum evidence: a prespecified causal graph, temporal ordering, measured overlap, at least ten independent outcome events per effective parameter or regularised alternative, and no critical unmeasured confounder falsely labelled controlled. Method: use design-stage restriction or matching, prespecified adjustment sets, flexible outcome models and negative controls; assess overlap and collider risk. Uncertainty: report residual imbalance, uncertainty, sensitivity to unmeasured confounding and how estimates change across defensible adjustment sets. Safe failure: make an association-only claim, narrow the population or state that confounding cannot be controlled.

ANSWER BOUNDARY

Do not add every available variable, adjust for mediators or colliders without justification, infer individual behaviour, or present an adjusted association as automatically causal.

WHY THIS IS USEFUL

Helps the team decide whether covariate adjustment is credible or whether the study should remain descriptive or be redesigned.

Category 5: Methods, baselines, confounding, bias and uncertainty

RD-045

STAKEHOLDER QUESTION

CATEGORY 5

How could sensor availability, room selection, access restrictions or exclusion rules make the analysed sample unrepresentative?

SENSORS AND TELEMETRY

Proposed evidence: Use coverage, health, permission and mapping metadata for both included and excluded devices, rooms and periods; no additional occupant sensing is needed. Spatial coverage and limits: Describe the full eligible building scope and all selected or unavailable units, including uninstrumented, inaccessible and low-uptime strata. Temporal coverage and aggregation: Create weekly inclusion and exposure summaries and preserve the exact reason and time each unit enters or leaves the sample. Record-specific quality check: verify mapping, calibration status, clocks, uptime, gaps, drift and provenance for selected and non-selected evidence so quality filters do not hide systematic exclusion. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: current space and facilities owners; sensor-platform and access owners; ethics and governance owners; principal investigator; and data steward. Precedence: the prespecified eligibility frame and current authoritative access status outrank the convenience sample that happens to be available. The May 2021 floor plan may support historical orientation only; current owner records and competent confirmation govern current use, access, services, hazards and closures. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: selection and permission states must be effective-dated throughout collection. Conflict rule: preserve denied, unhealthy and ambiguous units as named frame categories and do not silently remove them.

ANALYSIS REQUIRED

Operation: comparison plus estimate. Decision interval: at recruitment or site selection, monthly during collection and before population claims. Minimum evidence: a complete eligibility frame, inclusion probabilities or reasons, key structural variables for included and excluded units, and outcome-independent selection rules. Method: compare distributions, fit selection models or weights where support exists, and calculate worst-case bounds for missing strata. Uncertainty: report standardised differences, effective sample size after weighting and uncertainty under unobserved selection. Safe failure: limit inference to the observed sample, target a narrower population or collect evidence in underrepresented strata.

ANSWER BOUNDARY

Do not call a convenient instrumented sample representative, weight beyond common support, hide permission-related exclusions, or infer characteristics of unobserved rooms or users.

WHY THIS IS USEFUL

Makes coverage bias visible and supports a choice between rebalancing collection and narrowing the population claim.

RD-046

STAKEHOLDER QUESTION

CATEGORY 5

How should an anomaly detector be evaluated when true faults or unusual events are rare, labels are imperfect and normal operation changes?

SENSORS AND TELEMETRY

Proposed evidence: Use the approved environmental, BMS, metering or platform-health streams plus owner-confirmed event records and deliberately constructed safe test cases; do not create hazardous conditions or infer personal behaviour. Spatial coverage and limits: Cover multiple verified assets, rooms and operating regimes, with separate training and future or asset-held-out evaluation periods. Temporal coverage and aggregation: Retain native event timing and predeclared feature windows; score continuously but aggregate alerts by a defined incident or episode rule. Record-specific quality check: verify mapping, calibration relevance, clock alignment, uptime, gaps, drift, label timing and model-input provenance. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: facilities and competent technical owners; BMS or platform owners; maintenance event owner; model owner; principal investigator; and data steward. Precedence: competent-owner confirmation outranks model scores; absence of a logged fault is not proof that an alert is false, and an anomaly is not a fault label. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: label definitions and operating regimes must be effective for each event. Conflict rule: retain uncertain labels and score them separately rather than forcing binary truth.

ANALYSIS REQUIRED

Operation: forecast plus comparison; no operational diagnosis. Decision interval: during model development, before deployment and after regime or sensor change. Minimum evidence: an untouched time-or asset-held-out set, at least 30 confirmed events or a justified rare-event design, realistic normal exposure, defined alert burden and label uncertainty. Method: compare simple statistical and persistence baselines with the proposed model; use event-level precision, recall, time-to-detection, false alerts per asset-day and calibration. Uncertainty: report 95% intervals, prevalence sensitivity, performance by regime and degradation under label, drift and missing-data uncertainty. Safe failure: keep the model exploratory, require human triage, collect labels or decline fault-detection claims.

ANSWER BOUNDARY

Do not treat an anomaly score as a confirmed fault, optimise only for overall accuracy, use post-event information at prediction time or deploy alerts from a model whose false-alarm burden and drift response are unknown.

WHY THIS IS USEFUL

Shows whether the detector adds value beyond simple baselines and whether alert burden and uncertainty are acceptable for a bounded research deployment.

Category 5: Methods, baselines, confounding, bias and uncertainty

RD-047

STAKEHOLDER QUESTION

CATEGORY 5

Across which seasons, timetable regimes and operational states does a forecast remain calibrated, and how far ahead is it useful?

SENSORS AND TELEMETRY

Proposed evidence: Use approved historical and near-real-time outcome streams with only covariates available by forecast issue time, such as published weather forecasts, calendar features and known control schedules. Spatial coverage and limits: Cover the verified target rooms or systems and hold out complete future periods and, where possible, unseen units. Temporal coverage and aggregation: Retain native data, define issue time and horizon, and create fixed training, validation and rolling test origins; prevent late-arriving or future data from entering features. Record-specific quality check: verify mapping, calibration continuity, clocks, source latency, uptime, gaps, drift and feature-availability provenance at each forecast origin. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: source-system, weather, timetable, facilities and sensor owners; model owner; principal investigator; and data steward. Precedence: data actually available at issue time outrank retrospectively complete stores; current operating state and owner-confirmed changes outrank historical regularity. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: forecast inputs, mappings and schedules must be valid at every origin. Conflict rule: mark origins with unavailable or conflicting inputs as no-forecast rather than backfilling them with later knowledge.

ANALYSIS REQUIRED

Operation: forecast plus comparison. Decision interval: at each candidate horizon, before deployment and at monthly recalibration checks. Minimum evidence: at least one full seasonal cycle where seasonality is claimed or a narrower scope, rolling origins, a naive persistence and seasonal baseline, and enough events for calibration assessment. Method: use blocked rolling evaluation, probabilistic forecasts, calibration curves, proper scoring rules and performance by regime and horizon. Uncertainty: report prediction intervals, coverage, sharpness, calibration error, gain over the baseline and uncertainty under drift and missing inputs. Safe failure: issue no forecast, shorten the horizon, revert to the baseline or restrict to supported regimes.

ANSWER BOUNDARY

Do not use future information, present point predictions without uncertainty, promise accuracy after unobserved operational change, or let forecasts control systems without separate authority.

WHY THIS IS USEFUL

Identifies the horizons and regimes where forecasting improves decisions and where a simple baseline or no forecast is safer.

RD-048

STAKEHOLDER QUESTION

CATEGORY 5

What assumptions are needed for a causal interpretation, which can be strengthened by design, and which remain untestable?

SENSORS AND TELEMETRY

Proposed evidence: Use only evidence needed for exposure, outcome, measured common causes, timing and intervention assignment; add no intrusive sensing merely to make a causal claim possible. Spatial coverage and limits: Cover treated and comparison units, relevant shared systems and the full pre and post intervals with stable measurement. Temporal coverage and aggregation: Retain event and observation times at native precision, lock the temporal order and define exposure windows before outcome inspection. Record-specific quality check: verify mapping, calibration, clock alignment, uptime, gaps, drift and provenance equally across exposure groups. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: principal investigator; subject-matter and method experts; operational change owner; facilities and source owners; ethics and governance owners; and data steward. Precedence: documented intervention assignment and current operational facts outrank analyst inference; statistical adjustment cannot replace unmeasured design information. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: assumptions and source meanings must match the exact study period and target population. Conflict rule: retain competing causal structures and downgrade the claim when they imply materially different estimates.

ANALYSIS REQUIRED

Operation: estimate with causal interpretation only under an explicit identification strategy. Decision interval: before data inspection, after design diagnostics and before claim wording. Minimum evidence: a causal graph, temporal ordering, positivity, stable measurement, no material interference beyond the defined unit, a justified comparison and prespecified falsification tests. Method: prefer randomisation or phased assignment where feasible; otherwise use natural experiments, difference-in-differences, instrumental variables or adjustment only when assumptions fit the setting. Uncertainty: report assumption-specific estimates, 95% intervals, placebo and negative-control results and sensitivity to hidden bias. Safe failure: state an association, present multiple bounded explanations or declare the causal question unanswered.

ANSWER BOUNDARY

Do not present model adjustment as proof of causation, claim that measured covariates remove unmeasured confounding or use causal language when exposure timing, interference or counterfactual assumptions are unsupported.

WHY THIS IS USEFUL

Makes the causal claim conditional and testable and helps the team decide whether stronger design, narrower wording or a different question is needed.

Category 5: Methods, baselines, confounding, bias and uncertainty

RD-049

STAKEHOLDER QUESTION

CATEGORY 5

Which robustness checks, negative controls and alternative specifications could reveal dependence on one fragile analytical choice?

SENSORS AND TELEMETRY

Proposed evidence: Use the same verified study evidence plus prespecified negative-control variables, periods or units that should not respond under the proposed explanation; do not add arbitrary variables after seeing results. Spatial coverage and limits: Cover the complete primary sample and matched controls, with identical spatial and temporal inclusion rules. Temporal coverage and aggregation: Retain native data and generate a finite locked set of alternative windows, aggregations, exclusions and models; log every executed specification. Record-specific quality check: verify calibration, clocks, uptime, gaps, drift and provenance once and propagate identical quality rules across all specifications. Evidence gate: confirm modality, position, asset/zone mapping, commissioning, permission, calibration or reference evidence, UTC offset, cadence, uptime, gaps and backfill or resampling, drift, and device, firmware, configuration and raw-to-derived lineage; quarantine failed intervals.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: principal investigator; method lead; source owners; subject-matter expert; and data steward. Precedence: the prespecified robustness plan outranks post-result experimentation; negative controls require a defensible reason, not merely a null expectation. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: specifications must be frozen before headline interpretation and linked to the same dataset version. Conflict rule: show all planned results, including contradictory ones, and do not discard a check because it weakens the narrative.

ANALYSIS REQUIRED

Operation: comparison plus estimate. Decision interval: after the primary model is locked and before publication or operational communication. Minimum evidence: a primary specification, at least three substantively plausible alternatives, one or more negative controls where valid, identical data quality gates and a defined robustness criterion. Method: use specification curves, multiverse summaries, placebo dates or locations, leave-one-unit-out analysis and alternative missingness treatments. Uncertainty: report the distribution of estimates, sign and decision stability, control results and uncertainty without counting correlated specifications as independent evidence. Safe failure: downgrade the claim, identify the fragile assumption or report the result as unresolved.

ANSWER BOUNDARY

Do not search unlimited specifications, report only favourable checks, treat a failed negative control as irrelevant, or convert robustness frequency into a probability that the claim is true.

WHY THIS IS USEFUL

Shows whether the conclusion is stable enough to act on and identifies the exact modelling choice that needs stronger evidence.

RD-050

STAKEHOLDER QUESTION

CATEGORY 5

How should measurement error, sampling variation, missingness, model choice and transfer uncertainty be combined for the intended decision?

SENSORS AND TELEMETRY

Proposed evidence: Use calibrated measurement uncertainty, coverage and gap evidence, repeated samples across independent units, model outputs and target-context metadata; no additional sensing is justified unless one component dominates the decision. Spatial coverage and limits: Cover every evidence source contributing to the decision and the target units or contexts to which the recommendation may apply. Temporal coverage and aggregation: Calculate uncertainty at the declared decision interval and preserve component-specific inputs rather than collapsing quality into one score. Record-specific quality check: verify calibration and placement evidence, clocks, uptime, gaps, drift and provenance for every component before propagation. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: instrument and source owners; method lead; principal investigator; target decision owner; information governance where risk matters; and data steward. Precedence: owner-defined operational consequences and current evidence limits outrank a generic confidence threshold; competent professionals retain authority for safety and compliance decisions. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: loss, threshold and target-context assumptions must be current at the decision date. Conflict rule: present conflicting loss functions or thresholds separately and do not average away an owner disagreement.

ANALYSIS REQUIRED

Operation: calculation plus recommendation. Decision interval: at each major decision point and before communicating a recommended action. Minimum evidence: component uncertainty estimates, dependence assumptions, plausible model set, missingness scenarios, target-context overlap and an explicit consequence or loss table. Method: propagate uncertainty by simulation or Bayesian modelling, decompose variance where possible and perform decision or value-of-information analysis. Uncertainty: report total and component intervals, probability of crossing the decision threshold, expected regret and sensitivity to loss and transfer assumptions. Safe failure: recommend more evidence, a reversible pilot, no action or an explicitly cautious descriptive conclusion.

ANSWER BOUNDARY

Do not hide uncertainty in a traffic-light label, promise certainty, certify safety or compliance, or recommend operational control outside the responsible owner's authority.

WHY THIS IS USEFUL

Connects statistical uncertainty to a real choice and shows whether the next best action is adoption, a limited pilot, more evidence or no change.

Category 6: Privacy, ethics, consent and disclosure control

RD-051

STAKEHOLDER QUESTION

CATEGORY 6

What is the minimum data content, granularity and retention period that can answer the approved question?

SENSORS AND TELEMETRY

Proposed evidence: Use no person-identifying modality by default. Prefer room or system environmental telemetry, coarse anonymous counts and owner-held operational data, with fields and resolution limited to the stated estimand. Spatial coverage and limits: Restrict coverage to approved locations and periods; exclude sensitive rooms, identifiers and fine spatial traces unless separately justified and authorised. Temporal coverage and aggregation: Collect at the coarsest cadence that preserves the target signal, aggregate early where defensible and set deletion or irreversible reduction dates before collection. Record-specific quality check: verify source mapping and health, retain only clock, gap, drift and calibration metadata necessary for quality, and preserve provenance without exposing sensitive operational detail. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: principal investigator; data owners; information governance and security teams; ethics approval holder; records owner; and data steward. Precedence: approved purpose, minimisation and retention controls outrank analytical convenience; source owners may impose stricter limits. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: permissions and retention dates must be current and tied to the exact data specification. Conflict rule: adopt the stricter rule when approvals conflict and exclude data that cannot be justified field by field.

ANALYSIS REQUIRED

Operation: recommendation plus comparison. Decision interval: before collection, before linkage and at each retention checkpoint. Minimum evidence: a field-to-estimand justification, candidate resolutions, utility test, disclosure-risk assessment, storage and retention plan and deletion verification method. Method: compare analytical utility and risk across reduced datasets; test whether each field or resolution materially changes the prespecified result. Uncertainty: report utility loss intervals, residual risk and which conclusions become unsupported after minimisation. Safe failure: remove the field, aggregate further, shorten retention or decline the research question.

ANSWER BOUNDARY

Do not collect data for unspecified future use, retain maximum resolution by default, infer sensitive traits, or claim that de-identification eliminates all risk.

WHY THIS IS USEFUL

Creates a defensible data specification that reduces exposure, cost and governance burden without sacrificing the approved question.

RD-052

STAKEHOLDER QUESTION

CATEGORY 6

Which ethics, governance, data-owner and security approvals must be in force before collection, linkage, analysis or sharing begins?

SENSORS AND TELEMETRY

Proposed evidence: No sensing or extraction should begin for this question. Use a proposed data-flow diagram, modality and location specification, sample schema, access model and security-control evidence. Spatial coverage and limits: Describe every proposed source, location, recipient, linkage, environment and output route, including external collaborators and cloud services. Temporal coverage and aggregation: State collection cadence, aggregation, retention, refresh and deletion in the submission; do not test with real data before the relevant permissions exist. Record-specific quality check: include planned calibration, clock, uptime, gap, drift and provenance controls so approvals cover the actual evidence design rather than a vague sensor label. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: institutional ethics body; data and asset owners; information governance and security teams; facilities and safety owners; principal investigator; sponsor and data steward. Precedence: formal approvals and owner permissions outrank technical feasibility, informal agreement or prior project precedent. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: each approval must be current, purpose-specific, effective-dated and linked to the final data-flow specification. Conflict rule: stop the affected activity when requirements conflict and obtain a documented decision from the competent authority.

ANALYSIS REQUIRED

Operation: lookup plus recommendation. Decision interval: at concept design, before any pilot, and after material change in modality, linkage, purpose, recipient or risk. Minimum evidence: a complete data-flow and responsibility matrix, data classification, lawful or institutional basis, risk controls, participant route where applicable, retention, incident and sharing arrangements. Method: map every activity to required authority and evidence; test the proposed design against data minimisation, separation of duties and least privilege. Uncertainty: report unresolved requirements, dependencies and residual risk rather than a single approval label. Safe failure: redesign, defer or cancel the activity; use synthetic fixtures until permissions are complete.

ANSWER BOUNDARY

Do not begin collection or linkage because data are technically reachable, treat ethics approval as owner permission, or extend one approved purpose, population, modality, location or retention period to another without authority.

WHY THIS IS USEFUL

Gives the team a clear lawful and institutional route before real evidence is collected or linked.

Category 6: Privacy, ethics, consent and disclosure control

RD-053

STAKEHOLDER QUESTION

CATEGORY 6

When are participant information, consent or an opt-out route required, and how will participation remain genuinely voluntary?

SENSORS AND TELEMETRY

Proposed evidence: Prefer non-identifying room or system measures. Where participants are directly involved, use only the approved modality and visible study devices; avoid covert imaging, audio, wireless-device capture and biometric inference. Spatial coverage and limits: Limit collection to declared locations, populations and periods, with practical non-participation zones or alternative arrangements where the protocol requires them. Temporal coverage and aggregation: Run only during approved windows, display current notices and apply consent or opt-out state before collection or inclusion; aggregate as early as possible. Record-specific quality check: verify device mapping, calibration, clock alignment, uptime, gaps, drift and consent-state provenance without storing unnecessary identity links. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: ethics approval holder; principal investigator; institutional governance; building and space owners; participant-facing study team; and data steward. Precedence: the approved participant route and current notice or consent materials outrank researcher assumption; employment or study relationships require careful power assessment. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: materials and consent state must be valid at the time of collection and after protocol amendments. Conflict rule: exclude affected data when consent or notice evidence is missing, disputed or technically unenforceable.

ANALYSIS REQUIRED

Operation: lookup plus observation. Decision interval: before recruitment or collection and continuously during participant-facing activity. Minimum evidence: approved information and consent route, versioned notices, eligibility and withdrawal rules, technical enforcement tests and a record of coverage without unnecessary personal detail. Method: trace each collected record class to the applicable participant basis, test opt-out exclusions and assess whether non-participation is practical and non-penalising. Uncertainty: report notice reach limits, consent-state uncertainty and any sample bias introduced by opt-out or non-participation. Safe failure: pause collection, remove affected records or redesign to use non-participant aggregate evidence.

ANSWER BOUNDARY

Do not imply mandatory participation, conceal sensor capabilities, use consent for a different purpose, or identify people to enforce an aggregate research design.

WHY THIS IS USEFUL

Protects participants and gives the research team a credible way to collect only what people were accurately told and permitted.

RD-054

STAKEHOLDER QUESTION

CATEGORY 6

Could repeated room-level aggregates enable re-identification or reconstruction of individual routines when linked with external knowledge?

SENSORS AND TELEMETRY

Proposed evidence: Use the proposed aggregate dataset, release schedule and plausible auxiliary information; no new sensing is required to test disclosure risk. Spatial coverage and limits: Assess all rooms, time windows and rare operating periods, especially small groups, unusual schedules and stable repeated patterns. Temporal coverage and aggregation: Evaluate the exact proposed cadence, history length and update frequency; include cumulative risk from repeated releases, not only one snapshot. Record-specific quality check: verify spatial mapping and timestamp precision, source uptime and gaps, drift and provenance because irregular coverage can itself reveal routines. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: information governance and security teams; data owners; ethics approval holder; principal investigator; disclosure-control specialist; and data steward. Precedence: approved risk appetite and realistic attacker knowledge outrank claims that data are anonymous because names were removed. The May 2021 floor plan may support historical orientation only; current owner records and competent confirmation govern current use, access, services, hazards and closures. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: risk must be reassessed for each release, recipient, linkage context and cumulative history. Conflict rule: apply the stricter control when utility and disclosure assessments disagree or attacker knowledge is uncertain.

ANALYSIS REQUIRED

Operation: diagnosis of disclosure risk plus recommendation. Decision interval: before any sharing or dashboard release and after each material update. Minimum evidence: the exact proposed dataset, recipient and access model, plausible auxiliary data, repeated-release history, small-cell distribution and a defined acceptable-risk threshold. Method: conduct motivated-intruder scenarios, uniqueness and linkage analysis, differencing attacks and membership or attribute-inference tests where applicable. Uncertainty: report risk ranges, vulnerable strata, assumptions about attacker knowledge and residual risk after controls. Safe failure: aggregate further, delay or batch releases, suppress vulnerable cells, restrict access or withhold the dataset.

ANSWER BOUNDARY

Do not claim that aggregation guarantees anonymity, publish fine-grained occupancy-like traces, test re-identification using unauthorised external data or disclose a plausible individual routine.

WHY THIS IS USEFUL

Prevents apparently harmless aggregates from becoming a longitudinal record of identifiable routines.

Category 6: Privacy, ethics, consent and disclosure control

RD-055

STAKEHOLDER QUESTION

CATEGORY 6

How should small cells, rare events and extremes be protected without erasing the scientifically important pattern?

SENSORS AND TELEMETRY

Proposed evidence: Use the exact proposed table, figure or query output and its contributing counts, duration and uncertainty; no additional physical sensing is needed. Spatial coverage and limits: Cover every room, period, subgroup and event category in the output, including combinations created by filters or drill-down. Temporal coverage and aggregation: Assess both single and repeated releases at the declared aggregation interval; preserve suppressed values in the secure environment only when permitted. Record-specific quality check: verify mapping and time granularity, source uptime and gaps, drift flags and output lineage so suppressed and unsuppressed cells are reproducible. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: data owners; information governance; ethics approval holder; disclosure-control specialist; principal investigator; and data steward. Precedence: approved disclosure thresholds and contextual risk outrank a generic minimum count; rare operational events may also reveal sensitive building information. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: rules must be current for each recipient and release route. Conflict rule: use the more protective rule when count, duration, uniqueness or differencing risks conflict.

ANALYSIS REQUIRED

Operation: calculation plus recommendation. Decision interval: before every external output and interactive dashboard configuration. Minimum evidence: cell counts and exposure, secondary-disclosure paths, repeated-release history, plausible auxiliary knowledge, scientific utility target and approved risk threshold. Method: apply primary and complementary suppression, category combination, interval reporting, noise or controlled query budgets as appropriate; test differencing attacks. Uncertainty: report utility loss, residual disclosure risk and the number and type of protected cells without revealing their values. Safe failure: release a coarser summary, controlled-access output or no result for the vulnerable stratum.

ANSWER BOUNDARY

Do not reveal suppressed values through totals, filters or error bars, equate rarity with error, or remove important extremes without declaring the resulting scope limit.

WHY THIS IS USEFUL

Preserves valuable rare-event evidence where possible while preventing outputs from exposing individuals or sensitive operations.

RD-056

STAKEHOLDER QUESTION

CATEGORY 6

What additional privacy or security risk is created by linking two otherwise acceptable datasets, and is the linkage still proportionate?

SENSORS AND TELEMETRY

Proposed evidence: Use metadata and safe samples from the exact proposed sources, join keys, time and spatial resolution, recipient and output; do not perform the real linkage before approval. Spatial coverage and limits: Limit the assessment to approved locations and periods and identify whether combined data reveal routines, sensitive assets or operational dependencies absent from either source alone. Temporal coverage and aggregation: Model the declared linkage frequency, history length and refresh; include failed and ambiguous joins because they affect both privacy and scientific validity. Record-specific quality check: verify identifier semantics, clock alignment, uptime, gaps, drift flags and provenance needed for the join without expanding identity content. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: owners of every source; information governance and security teams; ethics approval holder; principal investigator; and data steward. Precedence: the combined-data risk classification and most restrictive source terms outrank the classification of each dataset alone. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: permissions must explicitly cover linkage purpose, keys, recipients, resolution, retention and outputs. Conflict rule: do not link when owner terms conflict; redesign with safer keys or a trusted linkage environment.

ANALYSIS REQUIRED

Operation: diagnosis of linkage risk plus recommendation. Decision interval: before linkage, after any added source or resolution change, and before sharing linked outputs. Minimum evidence: complete data-flow, join-key analysis, uniqueness and small-cell distribution, attacker model, source terms, linkage accuracy target and mitigation options. Method: compare unlinked and linked identifiability, attribute inference, operational sensitivity and join error; assess trusted-third-party, pseudonymous and aggregate alternatives. Uncertainty: report incremental risk, linkage-error intervals, vulnerable combinations and residual risk after controls. Safe failure: do not link, use coarser or one-time linkage, keep analysis inside a trusted environment or release only aggregate results.

ANSWER BOUNDARY

Do not assume two low-risk datasets remain low risk when combined, use personal identifiers for convenience, or let a technically successful join override source permissions.

WHY THIS IS USEFUL

Shows whether the scientific gain from linkage justifies the new risk and which architecture keeps that risk proportionate.

Category 6: Privacy, ethics, consent and disclosure control

RD-057

STAKEHOLDER QUESTION

CATEGORY 6

Are images, intelligible audio, wireless-device signals or other high-risk modalities necessary, or can lower-risk evidence answer the question?

SENSORS AND TELEMETRY

Proposed evidence: Default to no collection of identifiable images, intelligible audio, MAC addresses, device fingerprints or biometric signals. Compare safer alternatives such as non-recording sound level, low-resolution thermal count without storage, doorway beam counts or room-system telemetry. Spatial coverage and limits: Any exceptional modality must be confined to approved locations, fields of view or radio scope, exclude adjacent spaces and have visible safeguards and immediate reduction where feasible. Temporal coverage and aggregation: Collect only during approved windows at the minimum cadence and retention; disable raw capture when derived anonymous counts meet the purpose. Record-specific quality check: verify physical masking, calibration, clock alignment, uptime, gaps, drift, edge-reduction behaviour and secure provenance; independently test that raw high-risk content is not retained. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: ethics body; information governance and security teams; building and space owners; competent safety owner; modality specialist; principal investigator; and data steward. Precedence: formal necessity and proportionality approval outrank technical availability; any bystander, neighbouring-space or wireless scope restriction is authoritative. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: approval, signage, masking and retention must be current throughout collection. Conflict rule: stop collection immediately when masking, scope, notice, storage or deletion controls disagree or fail.

ANALYSIS REQUIRED

Operation: comparison plus recommendation. Decision interval: before procurement or deployment and after any capability or location change. Minimum evidence: a precise estimand, alternatives table, utility evidence, privacy and security threat model, technical safeguard tests, retention and deletion plan and explicit authority. Method: compare scientific utility, error, operational burden and residual risk of each modality; test raw-data leakage and adversarial reconstruction of derived outputs. Uncertainty: report uncertainty in utility and risk, safeguard failure probability and unsupported scenarios. Safe failure: choose the safer modality, collect no data, redesign the question or limit work to controlled laboratory evidence.

ANSWER BOUNDARY

Do not collect identifiable images, intelligible speech, device identifiers or biometric-like signals for convenience, infer identity or sensitive traits, or retain raw high-risk content after a safer derived measure is approved.

WHY THIS IS USEFUL

Forces a necessity test before high-risk sensing and usually directs the study toward simpler, more proportionate evidence.

RD-058

STAKEHOLDER QUESTION

CATEGORY 6

How will withdrawal, deletion, correction or objection be applied after participant-linked data have contributed to derived data or models?

SENSORS AND TELEMETRY

Proposed evidence: No new sensing is required. Use consent or participant-key records held separately, data and model lineage, snapshot manifests, deletion markers and output inventories. Spatial coverage and limits: Cover every store, backup, derived table, feature, model and shared copy that may contain or depend on the affected contribution, without exposing identity to analysts who do not need it. Temporal coverage and aggregation: Apply requests according to the approved time and scope rules; record receipt, effective action and any lawful or technical limitation in an access-controlled ledger. Record-specific quality check: verify identifier separation, clocked lineage, source copies, gaps in propagation and provenance of deletion or retraining actions. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: ethics approval holder; information governance and records teams; data and repository owners; principal investigator; model owner; and data steward. Precedence: the approved participant rights route and institutional decision outrank analyst preference; legal or ethical retention obligations may limit deletion and must be explained accurately. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: request status and applicable protocol must be current at the action date. Conflict rule: escalate conflicting obligations to the competent authority and freeze further use until scope is resolved.

ANALYSIS REQUIRED

Operation: lookup plus diagnosis of dependency. Decision interval: when a request is received and before the next analysis or release. Minimum evidence: a controlled participant-to-source key, complete lineage graph, snapshot and sharing inventory, approved action rule and verification evidence. Method: trace affected artefacts, execute deletion, exclusion or retraining where required, and compare outputs before and after when scientific interpretation may change. Uncertainty: report completion confidence, residual copies, model influence uncertainty and resulting scope limitations to authorised parties. Safe failure: pause use or sharing, withdraw affected outputs or state that the requested action cannot be verified yet.

ANSWER BOUNDARY

Do not expose participant identity broadly, promise deletion that backups or published aggregates cannot satisfy, conceal limitations, or continue use while a valid request remains unresolved.

WHY THIS IS USEFUL

Gives the team an executable rights pathway and prevents participant obligations from becoming impossible once data enter complex derivatives.

Category 6: Privacy, ethics, consent and disclosure control

RD-059

STAKEHOLDER QUESTION

CATEGORY 6

What immediate containment and notification steps apply if unauthorised access, unexpected sensitive content or disclosure-control failure is detected?

SENSORS AND TELEMETRY

Proposed evidence: Use access, download, query, export and repository telemetry; data-loss alerts; content-classification flags; and immutable security event records. Do not inspect more sensitive content than needed to contain the event. Spatial coverage and limits: Cover the affected systems, datasets, users, recipients and downstream copies; maintain separation between security evidence and scientific content. Temporal coverage and aggregation: Capture event time, detection time, containment actions and affected intervals; preserve logs at their approved resolution and retention. Record-specific quality check: verify clock alignment across security and data systems, log completeness, uptime and gaps, tamper evidence and event provenance. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: information security and governance incident owners; system and data owners; ethics approval holder; principal investigator; records owner; and data steward. Precedence: the institutional incident route and competent security decisions outrank researcher preference; containment may temporarily override analytical availability. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: contacts, severity criteria and response authority must be current and tested. Conflict rule: treat conflicting scope estimates conservatively until the incident owner resolves them.

ANALYSIS REQUIRED

Operation: diagnosis of incident scope plus recommendation to authorised owners. Decision interval: immediately on detection and at defined containment, eradication and recovery checkpoints. Minimum evidence: verified alerts or logs, affected asset and data inventory, access and export records, source clock bounds, owner contacts and an approved incident plan. Method: construct an event timeline, bound affected records and recipients, preserve evidence and test containment; assess research-validity implications separately. Uncertainty: report scope ranges, log gaps, confidence and residual exposure only to authorised recipients. Safe failure: stop access and sharing, isolate systems, preserve evidence and defer scientific work until competent owners permit resumption.

ANSWER BOUNDARY

Do not continue analysis on suspect copies, conceal the event, contact affected people without institutional direction or make legal, disciplinary or attribution claims from technical indicators alone.

WHY THIS IS USEFUL

Provides a safe failure route that protects people and evidence while allowing authorised owners to contain the event and decide when research may resume.

RD-060

STAKEHOLDER QUESTION

CATEGORY 6

Do the planned tables, figures, models and demonstrations pass disclosure control for their exact audience and release channel?

SENSORS AND TELEMETRY

Proposed evidence: Use the exact final outputs, underlying cell counts and lineage, model artefacts, dashboard filter combinations and realistic auxiliary information; no new sensing is required. Spatial coverage and limits: Assess all spatial, temporal and categorical breakdowns, including downloadable data, hover text, cached queries and repeated releases. Temporal coverage and aggregation: Test static and interactive outputs at the final resolution and update frequency; include cumulative and differencing attacks across versions. Record-specific quality check: verify mapping and timestamp granularity, source gaps and drift flags, output provenance, suppression consistency and removal of secrets or sensitive metadata. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: data owners; information governance and security teams; ethics approval holder; publication or communications owner; principal investigator; and data steward. Precedence: approved audience, licence, disclosure threshold and source restrictions outrank the convenience of an open link or live demonstration. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: clearance must correspond to the exact artefact version, recipient and channel. Conflict rule: withhold the output when scientific and disclosure decisions conflict until the competent owner agrees a safe form.

ANALYSIS REQUIRED

Operation: diagnosis of disclosure risk plus recommendation. Decision interval: before submission, presentation, repository deposit, demonstration and each material update. Minimum evidence: final artefact hashes, audience and channel, cell and uniqueness statistics, model extraction and inversion checks where relevant, secret scan and release permissions. Method: apply motivated-intruder testing, small-cell and differencing checks, model leakage tests, metadata inspection and independent assurance sampling. Uncertainty: report residual risk, withheld components, utility loss and limits on reproduction without revealing protected content. Safe failure: aggregate, redact, restrict access, replace with synthetic fixtures or do not release.

ANSWER BOUNDARY

Do not assume a figure is safe because raw data are absent, expose sensitive mappings, credentials or rare traces, or claim open availability when controls restrict use.

WHY THIS IS USEFUL

Determines whether each output can be released as planned, must be aggregated or access-controlled, or should be withheld because the disclosure risk cannot be reduced proportionately.

Category 7: Reproducibility, code, metadata and secure sharing

RD-061

STAKEHOLDER QUESTION

CATEGORY 7

Can the exact extraction be repeated from a frozen query, source snapshot and permission state without relying on analyst memory?

SENSORS AND TELEMETRY

Proposed evidence: No new physical sensing is required. Use version-controlled extraction code, parameter files, source identifiers, query plans, permissioned snapshots, checksums and row-count telemetry. Spatial coverage and limits: Cover every source, table, partition, location filter and time interval used by the study, including external files and manual imports. Temporal coverage and aggregation: Record query issue time, retrieval time, source watermark and any allowed lateness; keep a small approved test fixture for deterministic checks. Record-specific quality check: verify source mapping, calibration and clock fields are requested, quality flags for uptime, gaps and drift are retained, and extraction provenance is immutable. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: source and database owners; platform administrator; repository owner; principal investigator; code owner; and data steward. Precedence: the permissioned source snapshot and owner-issued schema outrank a local copy or remembered query; current access limits may prevent later rerun and must be recorded. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: queries, credentials route and source versions must be valid at extraction time and retained without secrets. Conflict rule: fail the rerun when source versions, parameters or row counts conflict beyond declared tolerances.

ANALYSIS REQUIRED

Operation: lookup plus comparison. Decision interval: for every frozen dataset and before release. Minimum evidence: versioned query and parameters, source and permission identifiers, expected schema and counts, hashes, environment identifier and a successful rerun on a clean account or service role. Method: execute the extraction twice or against a frozen snapshot, compare schemas, row identities, counts and hashes, and explain permitted nondeterminism such as late arrival. Uncertainty: report exact differences, unavailable sources and the reproducibility limit created by retention or permission expiry. Safe failure: preserve a controlled snapshot, publish only a specification and fixture, or label the extraction non-reproducible.

ANSWER BOUNDARY

Do not store credentials in code, rely on personal workspaces, overwrite the cited extract, or claim reproducibility when a crucial source or permission cannot be recreated.

WHY THIS IS USEFUL

Creates a defensible starting point for every downstream result and makes extraction failure distinguishable from analytical disagreement.

RD-062

STAKEHOLDER QUESTION

CATEGORY 7

Is the metadata sufficient for a researcher unfamiliar with Abacws to interpret every field, quality flag, location and time value correctly?

SENSORS AND TELEMETRY

Proposed evidence: No new sensing is needed. Use source schemas, device and asset metadata, codebooks, sample records, quality-flag logic, calibration and mapping registers and access conditions. Spatial coverage and limits: Cover every raw and derived field, device class, location level, schema era and data-quality state included in the shared or preserved dataset. Temporal coverage and aggregation: Document native cadence and every aggregate interval, effective dates and time semantics; include examples that do not expose restricted values. Record-specific quality check: verify calibration and unit references, clock fields, uptime and gap definitions, drift flags and raw-to-derived provenance for 100% of selected variables. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: source, schema, sensor, asset and space owners; information governance; code and model owners; principal investigator; and data steward. Precedence: owner-issued definitions and effective-dated mappings outrank analyst shorthand; public metadata must still obey security and location restrictions. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: metadata must match the exact released dataset and be updated when code, schema, mapping or licence changes. Conflict rule: mark unresolved fields unknown and block scientific use rather than fill them with plausible prose.

ANALYSIS REQUIRED

Operation: lookup plus diagnosis of metadata completeness. Decision interval: before internal handover, sharing and preservation. Minimum evidence: a mandatory metadata profile covering meaning, unit, modality, precision, cadence, aggregation, missingness, quality flags, time semantics, location scope, owner, permission, provenance and known limitations. Method: run machine-readable schema validation, cross-reference identifiers, test examples and ask an independent researcher to answer a fixed interpretation checklist. Uncertainty: report completeness percentages, ambiguous fields and the share of analyses affected by each gap. Safe failure: withhold the field, provide metadata-only remediation or restrict reuse to the originating team.

ANSWER BOUNDARY

Do not infer current room meaning from the historical plan, publish sensitive asset details, use undefined acronyms or claim a dataset is self-describing because column names exist.

WHY THIS IS USEFUL

Enables safe, accurate reuse and reduces the chance that future analysts reproduce code while misunderstanding the evidence.

Category 7: Reproducibility, code, metadata and secure sharing

RD-063

STAKEHOLDER QUESTION

CATEGORY 7

Can the computational environment be recreated with the same dependencies, system libraries and relevant hardware assumptions?

SENSORS AND TELEMETRY

Proposed evidence: No physical sensing is required. Use pinned dependency files, container or environment definitions, base-image digest, system-library inventory, hardware and accelerator metadata and a deterministic smoke-test dataset. Spatial coverage and limits: Cover every environment used for extraction, feature generation, modelling and figure creation, including secure and external collaborator settings. Temporal coverage and aggregation: Capture the environment at each release and execute periodic clean builds before software registries or packages disappear. Record-specific quality check: verify that data-quality fields for clocks, uptime, gaps, drift and provenance survive environment rebuilds and are not altered by locale or library defaults. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: code and environment owners; institutional computing and security teams; package and container registry owners; principal investigator; and data steward. Precedence: the frozen environment digest and institutional security policy outrank a floating latest tag or an analyst laptop; supported hardware claims require actual testing. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: images and dependencies must be available and permitted at the build date, with vulnerability status recorded. Conflict rule: use the approved reference environment when platforms differ and do not weaken security controls merely to reproduce an old stack.

ANALYSIS REQUIRED

Operation: comparison plus diagnosis of computational reproducibility. Decision interval: in continuous integration, before release and at scheduled preservation checks. Minimum evidence: pinned direct and transitive dependencies, system and locale settings, hardware assumptions, build instructions, reference outputs and a clean runner. Method: build from scratch, run the full smoke path and compare outputs within declared numerical tolerances across supported platforms. Uncertainty: report build failures, platform-specific differences, nondeterminism and unsupported dependencies. Safe failure: provide one supported reference environment, archive executable artefacts where permitted or narrow the reproducibility claim.

ANSWER BOUNDARY

Do not package secrets or restricted data, use mutable tags, promise indefinite executability, ignore critical vulnerabilities or claim platform independence without evidence.

WHY THIS IS USEFUL

Reduces dependency decay and helps collaborators separate software-environment differences from scientific differences.

RD-064

STAKEHOLDER QUESTION

CATEGORY 7

Which pipeline stages are stochastic or hardware-sensitive, and how stable are findings across seeds and execution conditions?

SENSORS AND TELEMETRY

Proposed evidence: No new sensing is required. Use fixed input snapshots, random seed logs, deterministic-algorithm settings, thread and accelerator configuration, training telemetry and output hashes. Spatial coverage and limits: Cover all stochastic stages, including data splitting, sampling, optimisation, hyperparameter search and approximate numerical routines. Temporal coverage and aggregation: Repeat complete runs across prespecified seeds and, where relevant, supported CPU or accelerator environments; preserve per-run parameters and timing. Record-specific quality check: verify that source mappings, clock fields, uptime, gaps, drift flags and provenance are identical across runs so input change is not mistaken for computational variation. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: model and code owners; computing-platform and security teams; repository owner; principal investigator; and data steward. Precedence: frozen inputs and configuration outrank notebook state; hardware-specific behaviour must be reported rather than normalised away without evidence. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: seed and platform metadata must be captured for every reported run. Conflict rule: stop comparison when inputs, code or environment differ beyond the intended factor.

ANALYSIS REQUIRED

Operation: estimate and comparison. Decision interval: during model selection and before final reporting. Minimum evidence: at least 20 seeds for materially stochastic headline metrics or a justified convergence study, fixed splits, identical inputs and two hardware conditions where sensitivity is plausible. Method: estimate between-run and between-platform variance, rank stability and probability of changing the scientific or operational conclusion. Uncertainty: report distributions and 95% intervals, worst-run behaviour, numerical tolerances and the share of total uncertainty due to computation. Safe failure: use a simpler stable method, increase repeated runs, fix a supported platform or state that the result is computationally fragile.

ANSWER BOUNDARY

Do not report the best seed, conceal nondeterminism, tune on test outcomes, or imply bitwise identity where only scientific equivalence was tested.

WHY THIS IS USEFUL

Shows whether a reported gain is a stable property of the method or an artefact of random initialisation or hardware.

Category 7: Reproducibility, code, metadata and secure sharing

RD-065

STAKEHOLDER QUESTION

CATEGORY 7

Which exact data, schema, mapping, code, environment and model versions produced each table, figure and reported number?

SENSORS AND TELEMETRY

Proposed evidence: No new sensing is required. Use figure and table manifests, data snapshot hashes, query and transformation identifiers, mapping and calibration versions, code commits, environment digests, model cards and parameter files. Spatial coverage and limits: Cover every output, including supplementary tables, dashboard exports and numbers copied into narrative text. Temporal coverage and aggregation: Generate a manifest at output creation and preserve it with the artefact; use stable result identifiers in captions and source documents. Record-specific quality check: verify that quality metadata for calibration, clocks, uptime, gaps and drift are linked to the exact input rows and not lost in summarisation. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: data and repository owners; code and model owners; calibration and mapping stewards; principal investigator; publication owner; and data steward. Precedence: machine-generated manifests and immutable hashes outrank filenames or manual notes; source permissions govern what manifest detail may be public. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: manifests must be created from the exact final artefact and frozen before submission or release. Conflict rule: withhold or regenerate any output whose manifest conflicts with the current file or cited result.

ANALYSIS REQUIRED

Operation: lookup plus observation. Decision interval: whenever an output is generated and before it is cited, submitted or presented. Minimum evidence: unique result identifier, hashes for artefact and inputs, schema and mapping versions, code commit, environment, parameters, model and execution time, plus a successful trace to raw evidence. Method: automatically bind manifests to outputs, sample-recompute reported values and compare captioned sample sizes and intervals with the underlying result object. Uncertainty: report unmatched outputs, stale manifests and numerical discrepancies with exact scope. Safe failure: remove the output, recreate it from a verified run or label it non-citable.

ANSWER BOUNDARY

Do not copy numbers between versions without trace, edit figures manually without preserving source data, expose restricted paths, or cite a model name without its exact version.

WHY THIS IS USEFUL

Makes every result independently traceable and sharply reduces errors when manuscripts, presentations and repositories evolve.

RD-066

STAKEHOLDER QUESTION

CATEGORY 7

Which automated checks must stop the pipeline when data, mappings, methods or outputs violate a critical expectation?

SENSORS AND TELEMETRY

Proposed evidence: No new physical sensing is required. Use test fixtures, source health and schema telemetry, calibration and clock status, expected sample manifests, mapping registries and output invariants. Spatial coverage and limits: Cover every critical source, transformation and reported output, including negative cases that should fail. Temporal coverage and aggregation: Run checks at ingestion, before feature generation, before modelling and before release; retain test results and exact input hashes. Record-specific quality check: verify mapping, calibration validity, UTC offset, uptime, gaps, drift, schema and provenance through fail-fast assertions rather than warning-only dashboards. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: source and platform owners; code and model owners; method lead; information security where relevant; principal investigator; and data steward. Precedence: owner-defined critical limits and the approved analysis specification outrank developer convenience; a warning may not substitute for a required stop. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: tests and limits must be versioned with the method and source era. Conflict rule: block promotion when code, owner declarations and observed evidence disagree on a critical condition.

ANALYSIS REQUIRED

Operation: diagnosis of pipeline validity plus recommendation. Decision interval: on every code change and analytic run. Minimum evidence: unit, integration and end-to-end tests; known-good and known-bad fixtures; expected schema, range, completeness, mapping, timing, leakage and output checks; and declared stop conditions. Method: run tests in continuous integration and the secure environment, inject controlled failures, test reproducible seeds and compare outputs with golden tolerances. Uncertainty: report coverage of critical paths, false-pass evidence, flaky tests and untested assumptions. Safe failure: stop the run, isolate an unaffected subset or produce only a quality report with no scientific result.

ANSWER BOUNDARY

Do not downgrade a critical failure to a warning, let validation tests learn from final outcomes, expose restricted fixtures or claim correctness merely because the pipeline completed.

WHY THIS IS USEFUL

Converts evidence requirements into enforceable gates and prevents known-invalid inputs from quietly reaching publication.

Category 7: Reproducibility, code, metadata and secure sharing

RD-067

STAKEHOLDER QUESTION

CATEGORY 7

Which secure analysis environment is proportionate to the data sensitivity, scale and collaboration pattern?

SENSORS AND TELEMETRY

Proposed evidence: No new sensing is required. Use data classification, identity and access telemetry, encryption and key-management evidence, logging, network and export controls, recovery tests and compute-capacity measurements. Spatial coverage and limits: Assess every storage, compute, notebook, model service, repository and transfer route used by the team and collaborators. Temporal coverage and aggregation: Evaluate controls before provisioning and at least annually or after material data, threat or architecture change; retain security events at approved resolution. Record-specific quality check: verify source provenance, clock alignment of security logs, uptime and gaps in monitoring, and configuration drift of access and export controls. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: information security and governance teams; data and system owners; institutional computing; ethics approval holder; principal investigator; sponsor and data steward. Precedence: institutional classification and competent security approval outrank researcher preference or consumer-service convenience. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: approvals, user lists, threat model and controls must be current for the exact dataset and collaborators. Conflict rule: select the more restrictive supported environment when owner or control evidence conflicts.

ANALYSIS REQUIRED

Operation: diagnosis of security posture plus recommendation. Decision interval: before provisioning, at access renewal and after material change. Minimum evidence: data classification, users and roles, scale and performance needs, threat model, retention, recovery, logging, export path, source terms and owner approval. Method: map threats to controls, test least privilege, encryption, logging, export restrictions, recovery and access removal, and compare enclave, code-to-data and controlled workspace options. Uncertainty: report residual risk, control-test confidence, capacity limits and unresolved critical findings. Safe failure: use a more restricted enclave, reduce data, use code-to-data or do not analyse the dataset.

ANSWER BOUNDARY

Do not upload restricted data to unapproved services, weaken controls for convenience, promise absolute security, or let the catalogue select an environment without competent authority.

WHY THIS IS USEFUL

Matches safeguards to real risk and gives collaborators a supported place to work without creating unmanaged copies.

RD-068

STAKEHOLDER QUESTION

CATEGORY 7

How can collaborators reproduce the work while staying within their own permissions, institutional controls and data-use terms?

SENSORS AND TELEMETRY

Proposed evidence: No new sensing is required. Use tiered data packages, metadata, code, safe fixtures, controlled queries, remote execution and result manifests tailored to each collaborator's approved access. Spatial coverage and limits: Cover only the variables, locations and intervals each recipient is permitted to use; keep restricted mappings and raw streams inside the approved environment. Temporal coverage and aggregation: Provide frozen versions and scheduled controlled refreshes rather than unbounded live access; retain access and export telemetry. Record-specific quality check: verify that calibration, clock, uptime, gap, drift and provenance metadata needed for scientific interpretation remain available at each access tier. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: data owners; collaborators' institutions; information governance and security teams; repository and platform owners; principal investigator; and data steward. Precedence: the most restrictive applicable source term and recipient control outrank a desire for identical local copies. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: agreements, user permissions, licences and environments must be current for every recipient and artefact. Conflict rule: pause sharing when recipient and source controls conflict; offer a safer metadata or code-to-data route.

ANALYSIS REQUIRED

Operation: comparison plus recommendation. Decision interval: before collaboration begins and at each new dataset or recipient. Minimum evidence: recipient-specific permission matrix, reproducibility target, data and compute need, source terms, safe package options, tested access and output checking. Method: compare open code plus fixture, controlled download, remote workspace and code-to-data designs on scientific utility, risk, cost and traceability. Uncertainty: report which claims can be reproduced at each tier, residual differences and access-dependent uncertainty. Safe failure: share metadata and code only, run collaborator code inside the enclave or state that independent reproduction is not currently possible.

ANSWER BOUNDARY

Do not send restricted copies by convenience, assume a collaboration agreement grants data rights, conceal access-dependent limits or promise identical reproduction across unequal evidence tiers.

WHY THIS IS USEFUL

Enables collaboration while making clear which parts can be independently reproduced and which require controlled execution.

Category 7: Reproducibility, code, metadata and secure sharing

RD-069

STAKEHOLDER QUESTION

CATEGORY 7

Can a synthetic or sampled dataset support code development without exposing real sensitive patterns or teaching the wrong data behaviour?

SENSORS AND TELEMETRY

Proposed evidence: No new sensing is required. Use schema-only mocks, generated time series, approved small samples or synthetic records created from aggregate constraints; keep real restricted benchmarks inside the secure environment. Spatial coverage and limits: Represent the field types, ranges, missingness, quality flags, mapping relationships and edge cases needed for development without copying rare real episodes. Temporal coverage and aggregation: Generate fixed seeds and versions, include native-like cadence and gaps, and define which temporal and cross-variable relationships are intentionally absent or approximate. Record-specific quality check: verify that calibration, clock, uptime, gap, drift and provenance fields are structurally represented and that no source record can be traced into the synthetic set. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: data owners; information governance and security teams; code owner; method lead; principal investigator; and data steward. Precedence: approved disclosure controls and source terms outrank development convenience; synthetic status does not automatically make data safe or scientifically valid. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: generator, constraints and risk assessment must match each release version. Conflict rule: withhold the dataset when privacy and utility tests disagree or rare records appear too similar to source data.

ANALYSIS REQUIRED

Operation: comparison plus diagnosis of privacy and utility. Decision interval: before distribution and after generator or source changes. Minimum evidence: target coding tasks, prohibited disclosures, exact candidate set, generator version, held-out restricted benchmark and prespecified utility and risk criteria. Method: test schema, constraints, edge cases, temporal and cross-variable behaviour, nearest-neighbour and membership risk, and software defect detection against the restricted benchmark. Uncertainty: report utility gaps, disclosure-risk bounds and tasks that still require controlled real-data testing. Safe failure: release schema-only fixtures, simplify synthetic relationships or keep all testing in the secure environment.

ANSWER BOUNDARY

Do not label data synthetic without disclosure testing, use real rare records as templates, assume statistical similarity preserves task validity or release a development fixture that reveals protected structure.

WHY THIS IS USEFUL

Supports safer software development while keeping the boundary between test convenience and scientific validity explicit.

RD-070

STAKEHOLDER QUESTION

CATEGORY 7

What must be preserved so the study remains interpretable and reproducible after active access, software services and project staff change?

SENSORS AND TELEMETRY

Proposed evidence: No new sensing is required. Preserve approved data snapshots or durable queries, metadata, schemas, mappings, calibration evidence, quality summaries, code, environments, models, manifests, permissions and decision records. Spatial coverage and limits: Cover every component needed to reproduce reported claims while minimising sensitive raw data and operational detail. Temporal coverage and aggregation: Create preservation packages at major releases and project closure; define retention, deletion and format-migration checkpoints. Record-specific quality check: verify hashes, clock and time semantics, uptime and gap summaries, drift and calibration history, and complete provenance before archival deposit. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: records and repository owners; data and source owners; information governance and security teams; principal investigator; sponsor; code and model owners; and data steward. Precedence: records, retention, licence and deletion obligations outrank a desire to keep everything; trusted repositories and controlled formats outrank personal storage. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: preservation permissions, formats and contact ownership must be current at deposit and rechecked at scheduled intervals. Conflict rule: preserve metadata and evidence of lawful deletion when raw data cannot be retained.

ANALYSIS REQUIRED

Operation: recommendation plus observation. Decision interval: at release, project closure and scheduled preservation checks. Minimum evidence: a claim-to-artefact inventory, retention and access authority, open-format or executable preservation plan, checksums, dependency risk assessment and named future custodian. Method: test restore on a clean environment, validate hashes and metadata, identify external-service dependencies and simulate loss of staff knowledge. Uncertainty: report recoverability, unavailable dependencies, retention limits and which claims remain reproducible at each future tier. Safe failure: preserve a reduced evidence package, deposit metadata and code only, or state the date beyond which reproduction is not assured.

ANSWER BOUNDARY

Do not retain restricted data indefinitely, rely on personal accounts or undocumented services, promise permanent execution, or discard provenance while keeping only final figures.

WHY THIS IS USEFUL

Protects the scientific value of the work and makes future limitations predictable rather than discovering them after access and expertise have disappeared.

Category 8: Interpretation, visualisation, publication and impact

RD-071

STAKEHOLDER QUESTION

CATEGORY 8

How should uncertainty, missingness and data quality be displayed so a figure does not imply more confidence or coverage than the evidence supports?

SENSORS AND TELEMETRY

Proposed evidence: Use the exact result data plus calibration uncertainty, sample counts, coverage, clock and mapping confidence, gap and drift flags and model uncertainty; no new sensing is required. Spatial coverage and limits: Represent every room, period or asset included in the visual and explicitly mark unsupported or incomparable strata. Temporal coverage and aggregation: Display uncertainty at the same interval as the estimate and show coverage over time; interactive views must retain these cues under filtering. Record-specific quality check: verify output provenance and that aggregation carries calibration, uptime, gap, drift and mapping-quality fields into the plotting dataset. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: data and source owners; method lead; visualisation author; principal investigator; information governance where disclosure risk applies; and data steward. Precedence: the frozen result object and quality metadata outrank a manually edited chart; source restrictions determine spatial and temporal detail. The May 2021 floor plan may support historical orientation only; current owner records and competent confirmation govern current use, access, services, hazards and closures. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: visuals must match the cited dataset and method version. Conflict rule: withhold or annotate strata when quality and result datasets disagree.

ANALYSIS REQUIRED

Operation: recommendation plus comparison. Decision interval: during figure design and before every presentation or release. Minimum evidence: point or distribution estimates, uncertainty intervals, sample and exposure counts, missingness and quality by stratum, disclosure check and a stated reading task. Method: compare interval, distribution and uncertainty-encoding designs; test whether users can identify unsupported periods, sample imbalance and decision ambiguity. Uncertainty: report interval meaning, coverage and quality thresholds, and interpretation error from a small structured comprehension test where practical. Safe failure: use a table, simplify the claim, remove unsupported strata or state that the evidence is too incomplete for a comparative visual.

ANSWER BOUNDARY

Do not truncate axes to exaggerate differences, connect across missing intervals, use colour alone for critical meaning, suppress low-quality strata or draw a smooth spatial surface where measurements are sparse.

WHY THIS IS USEFUL

Helps readers see both the result and the strength of evidence behind it, reducing confident but incorrect interpretation.

RD-072

STAKEHOLDER QUESTION

CATEGORY 8

Could aggregation across rooms, periods or operating states reverse, dilute or conceal the pattern in important strata?

SENSORS AND TELEMETRY

Proposed evidence: Use the verified analytical dataset with room, system, calendar, weather and operating-state strata retained at the minimum permitted detail; no person-level subgrouping is needed. Spatial coverage and limits: Cover all units contributing to the aggregate, including low-coverage and excluded strata, and retain weights and exposure time. Temporal coverage and aggregation: Compare native analytical intervals, prespecified strata and the published aggregate; preserve partial-window and missingness indicators. Record-specific quality check: verify calibration comparability, clocks, uptime, gaps, drift, mapping and provenance before attributing differences to aggregation. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: current space, facilities, BMS, timetable and sensor owners; method lead; principal investigator; and data steward. Precedence: current owner-confirmed strata and physical system boundaries outrank convenient grouping labels; the historical plan cannot define current room-use groups. The May 2021 floor plan may support historical orientation only; current owner records and competent confirmation govern current use, access, services, hazards and closures. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: strata must be effective for the measurement interval and frozen before interpretation. Conflict rule: show unresolved classification separately and do not allocate it to the group that strengthens the aggregate.

ANALYSIS REQUIRED

Operation: comparison plus estimate. Decision interval: before aggregation is selected and before headline interpretation. Minimum evidence: stratum definitions, counts and exposure, common outcome scale, adequate overlap, at least three independent units per reported stratum where possible and identical quality gates. Method: compare pooled, stratified, weighted and hierarchical estimates; test interactions and decomposition of within-and between-stratum patterns. Uncertainty: report heterogeneity intervals, weight sensitivity and whether sign or decision changes under defensible aggregation. Safe failure: publish stratified results, narrow the claim or state that no single building-wide summary is meaningful.

ANSWER BOUNDARY

Do not hide a reversal behind a building average, create strata after seeing outcomes, infer sensitive groups, or imply that a weighted average describes every room.

WHY THIS IS USEFUL

Prevents Simpson-type reversals and helps the team choose a summary that respects physically and operationally important differences.

Category 8: Interpretation, visualisation, publication and impact

RD-073

STAKEHOLDER QUESTION

CATEGORY 8

How should thresholds, targets and reference bands be labelled so readers can see their owner, effective date, averaging basis, applicability and uncertainty?

SENSORS AND TELEMETRY

Proposed evidence: Use the final measured or estimated values, their units and averaging intervals, instrument uncertainty and the exact candidate thresholds or bands; no new sensing is required. Spatial coverage and limits: Apply only to verified locations, modalities and conditions within the source scope; avoid extending a band from one space type or averaging period to another. Temporal coverage and aggregation: Display threshold effective dates and compare on compatible time windows; keep values near boundaries as uncertain classifications where measurement error overlaps the band. Record-specific quality check: verify calibration, clocks, uptime, gaps, drift, mapping and provenance for the values being compared. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: the competent owner of each threshold or target; facilities, health and safety or other professional authority where applicable; instrument owner; principal investigator; and data steward. Precedence: current owner-issued and scope-compatible references outrank generic guidance, old documents or research convenience; research comparisons do not create compliance authority. The May 2021 floor plan may support historical orientation only; current owner records and competent confirmation govern current use, access, services, hazards and closures. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: sources must be current at observation and publication and retain effective dates and version. Conflict rule: show applicable alternatives and source disagreement rather than choosing a rhetorically useful line.

ANALYSIS REQUIRED

Operation: lookup plus calculation. Decision interval: before visualisation, categorisation and publication. Minimum evidence: named source and owner, effective date, unit, averaging period, population or space scope, measurement uncertainty and intended descriptive or normative status. Method: build a threshold register, test unit and interval compatibility, calculate boundary ambiguity and compare plausible applicable sources. Uncertainty: report classification intervals, proportion near boundaries and sensitivity to measurement and source uncertainty. Safe failure: omit the band, display continuous values only or state that no applicable reference is confirmed.

ANSWER BOUNDARY

Do not present a research band as a statutory, clinical or safety limit, omit its averaging basis or effective date, or apply it to a modality, location or population outside the owner's stated scope.

WHY THIS IS USEFUL

Lets readers distinguish observed evidence from an externally owned reference and prevents a persuasive line on a chart from becoming an unsupported verdict.

RD-074

STAKEHOLDER QUESTION

CATEGORY 8

Which outputs should remain exploratory displays, and which have enough frozen evidence and method control to become fixed, citable figures?

SENSORS AND TELEMETRY

Proposed evidence: Use result manifests, data and method versions, dashboard query telemetry, quality gates, disclosure status and figure-generation code; no new sensing is required. Spatial coverage and limits: Cover every data source, filter, location and period available in the exploratory view and the exact subset proposed for citation. Temporal coverage and aggregation: Exploratory views may refresh under controlled labels; citable figures require a frozen snapshot, fixed query, method and retrieval date. Record-specific quality check: verify calibration, clock, uptime, gap, drift and mapping quality are frozen with the citable data and visibly signalled in exploratory views. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: data and platform owners; dashboard and code owners; principal investigator; publication owner; information governance; and data steward. Precedence: the frozen manifest, permissions and disclosure clearance outrank a screenshot or live URL; exploratory convenience does not confer citation stability. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: status must be reassessed after data, method, permission or dashboard change. Conflict rule: downgrade an output to exploratory when any critical version or gate conflicts.

ANALYSIS REQUIRED

Operation: lookup plus recommendation. Decision interval: at dashboard publication, figure freeze and every material update. Minimum evidence: explicit output purpose, frozen data and code identifiers, stable method, complete quality and uncertainty, disclosure clearance, caption and archival route. Method: apply a citable-output checklist, regenerate from a clean environment and compare the fixed figure with the live view at the freeze time. Uncertainty: report unresolved dynamic elements, refresh risk and the exact scope of citation stability. Safe failure: label the output exploratory, archive a fixed export or withhold it from evidentiary claims.

ANSWER BOUNDARY

Do not cite a live dashboard as immutable evidence, remove exploratory labels, manually edit the fixed figure without regenerating it, or imply that visual polish equals validation.

WHY THIS IS USEFUL

Separates rapid exploration from stable evidence and gives readers a durable object that can be traced and reproduced.

Category 8: Interpretation, visualisation, publication and impact

RD-075

STAKEHOLDER QUESTION

CATEGORY 8

Does every planned claim trace to its data, method, uncertainty, alternative explanations and a concrete answer boundary?

SENSORS AND TELEMETRY

Proposed evidence: No new sensing is required. Use the final manuscript or presentation claims, result identifiers, data and method manifests, uncertainty outputs, robustness checks, disclosure decisions and owner-confirmed context. Spatial coverage and limits: Cover every substantive headline and supporting claim, including figure captions, abstracts, press material and impact statements. Temporal coverage and aggregation: Freeze the claim set against the cited result version and retain any later wording change with its linked evidence. Record-specific quality check: verify that calibration, clock, uptime, gap, drift, mapping and provenance limitations from source evidence remain visible in the claim trace. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: principal investigator; method and subject-matter leads; data and source owners; publication or communications owner; information governance; and data steward. Precedence: the frozen evidence and competent-owner context outrank persuasive wording; source-owner authority limits operational, safety and compliance statements. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: claim traces must match the exact submitted or released version. Conflict rule: remove or downgrade claims whose evidence, permission or interpretation conflicts cannot be resolved.

ANALYSIS REQUIRED

Operation: lookup plus diagnosis of claim support. Decision interval: before submission, presentation and public communication. Minimum evidence: a claim-evidence matrix linking each claim to data snapshot, method, sample, uncertainty, robustness, alternative explanations, permitted scope and prohibited inference. Method: independently assurance-check all headline claims and a stratified sample of supporting claims; test whether wording changes when one key assumption fails. Uncertainty: report unsupported clauses, evidence gaps and the maximum defensible wording for each affected claim. Safe failure: remove the claim, label it exploratory or state the unresolved uncertainty directly.

ANSWER BOUNDARY

Do not present correlation as causation, infer people or sensitive traits, certify safety or compliance, hide uncertainty, or let fluent narrative substitute for evidence.

WHY THIS IS USEFUL

Provides a claim-by-claim gate for the paper, dataset note, presentation and stakeholder briefing, making unsupported wording easy to narrow before release.

RD-076

STAKEHOLDER QUESTION

CATEGORY 8

How should null, negative or inconclusive results distinguish precise absence, weak evidence and unresolved uncertainty?

SENSORS AND TELEMETRY

Proposed evidence: Use the final estimates, precision target, calibration and quality uncertainty, sample and exposure counts, missingness, model diagnostics and sensitivity results; no new sensing is required unless the study is clearly under-informative. Spatial coverage and limits: Cover the intended population and all prespecified strata, including unsupported periods and units that constrain interpretation. Temporal coverage and aggregation: Report the declared decision interval and minimum meaningful effect, not only a p-value at one aggregation or window. Record-specific quality check: verify mapping, calibration, clock, uptime, gaps, drift and provenance limitations are included in the uncertainty account. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: principal investigator; method lead; source owners; subject-matter expert; publication owner; and data steward. Precedence: the prespecified estimand and meaningful-effect threshold outrank post-result reframing; competent owners retain operational interpretation authority. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: thresholds and claim scope must be frozen before outcome inspection. Conflict rule: show conflicting method interpretations and avoid selecting the one that gives a cleaner conclusion.

ANALYSIS REQUIRED

Operation: estimate and recommendation. Decision interval: after the final locked analysis and before claim wording. Minimum evidence: effect estimate and interval, minimum meaningful effect, achieved independent information, missingness and bias sensitivity, model calibration and equivalence or non-inferiority margin where appropriate. Method: interpret confidence or credible intervals, equivalence tests, power or precision diagnostics and sensitivity bounds together; compare with simple baselines. Uncertainty: report the range of effects still compatible with evidence and the probability or confidence that meaningful effects are excluded under stated assumptions. Safe failure: state that the result is inconclusive, recommend more targeted evidence or limit the conclusion to the observed range.

ANSWER BOUNDARY

Do not claim no effect from non-significance, hide wide intervals, move the primary outcome after inspection or treat low-quality data as neutral evidence.

WHY THIS IS USEFUL

Separates a precise absence of a meaningful effect from an under-informative result, guiding publication wording and the decision to stop or collect targeted evidence.

Category 8: Interpretation, visualisation, publication and impact

RD-077

STAKEHOLDER QUESTION

CATEGORY 8

Which code, metadata, derived data and model artefacts can be released, under which licence, access condition and disclosure control?

SENSORS AND TELEMETRY

Proposed evidence: No new sensing is required. Use an item-level release inventory, source terms, ownership evidence, dependency licences, secret scans, disclosure tests, manifests and safe fixtures. Spatial coverage and limits: Cover every repository file, data object, model weight, configuration, notebook, figure source and documentation item proposed for deposit. Temporal coverage and aggregation: Assess the exact release version and update policy; frozen releases require hashes and a stated support horizon. Record-specific quality check: verify that calibration, clock, uptime, gap, drift, mapping and provenance metadata are sufficient for interpretation but do not expose sensitive operations. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: data, code, model and institutional intellectual-property owners; information governance and security teams; repository owner; principal investigator; funder where applicable; and data steward. Precedence: source rights, owner decisions and disclosure controls outrank an intended repository licence; one permissive component cannot relicence restricted evidence. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: licences, permissions and dependency terms must be current for the exact release. Conflict rule: withhold or separate items when rights or disclosure findings conflict and document the resulting reproduction limit.

ANALYSIS REQUIRED

Operation: lookup plus recommendation. Decision interval: before repository deposit and each release update. Minimum evidence: item-level owner, licence, permission, sensitivity, dependency rights, citation, support statement, secret scan, provenance and disclosure assessment. Method: use a release matrix, licence compatibility check, motivated-intruder test, model leakage test and clean-room packaging test. Uncertainty: report withheld components, residual risk, licence uncertainty and which results remain reproducible from the released package. Safe failure: release metadata, code and safe fixtures only, use controlled access or do not release the artefact.

ANSWER BOUNDARY

Do not publish without rights, assume a repository licence overrides source terms, expose credentials or sensitive mappings, or claim open reproducibility when key evidence remains restricted.

WHY THIS IS USEFUL

Enables the widest responsible reuse while making legal, security and scientific limitations explicit.

RD-078

STAKEHOLDER QUESTION

CATEGORY 8

How should findings be communicated to facilities teams, leaders and building users without overstating certainty or implying operational authority?

SENSORS AND TELEMETRY

Proposed evidence: Use the final verified results, quality and uncertainty summaries, current owner context, decision intervals and audience-specific disclosure controls; no new sensing is required for communication. Spatial coverage and limits: Tailor spatial detail to each audience and omit sensitive assets, small cells and unsupported room-level claims. Temporal coverage and aggregation: Use the exact frozen evidence date and clearly distinguish observation, calculation, estimate, forecast, candidate diagnosis and recommendation. Record-specific quality check: verify that calibration, clock, uptime, gap, drift, mapping and provenance limitations remain attached to every communicated metric. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: principal investigator; facilities and competent professional owners; communications owner; information governance; stakeholder representatives; and data steward. Precedence: current operational owners decide actions, safety, compliance and service meaning; research evidence may inform but cannot authorise or control. The May 2021 floor plan may support historical orientation only; current owner records and competent confirmation govern current use, access, services, hazards and closures. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: context and operational status must be confirmed at the communication date. Conflict rule: remove or reframe statements when owner context conflicts with the research interpretation.

ANALYSIS REQUIRED

Operation: recommendation plus comparison. Decision interval: before each briefing, report, dashboard or public statement. Minimum evidence: audience decision need, frozen claim-evidence matrix, uncertainty, alternatives, disclosure limits, named action owner and a plain-language boundary statement. Method: create layered summaries, compare interpretation across audiences and test whether readers distinguish findings, uncertainty and owner authority. Uncertainty: report known misunderstandings, omitted detail and confidence limits; retain technical annexes for authorised readers. Safe failure: communicate a narrower descriptive finding, defer to the owner or withhold an operational recommendation.

ANSWER BOUNDARY

Do not issue operational instructions, certify safety or compliance, promise savings or comfort, conceal uncertainty, or present a room-level aggregate as a judgement about individuals.

WHY THIS IS USEFUL

Helps stakeholders act on the right part of the evidence while preserving authority, uncertainty and privacy boundaries.

Category 8: Interpretation, visualisation, publication and impact

RD-079

STAKEHOLDER QUESTION

CATEGORY 8

How will the team determine whether the research changed a decision, method, data practice or building outcome rather than only producing outputs?

SENSORS AND TELEMETRY

Proposed evidence: Use prospective decision and engagement records, dated adoption evidence, method or policy artefact versions and, only where approved, aggregate outcome telemetry already justified by the impact question. Spatial coverage and limits: Cover the named decision owners, affected practice or system and an appropriate comparison or baseline; avoid tracking individual staff or users. Temporal coverage and aggregation: Define milestone dates and follow-up intervals before engagement; collect outcome evidence at the same prespecified aggregation before and after any adoption. Record-specific quality check: verify mapping, calibration, clock alignment, uptime, gaps, drift and provenance for any outcome telemetry, and preserve dated documentary evidence for non-sensor outcomes. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: decision and practice owners; facilities or organisational leaders; principal investigator; sponsor or funder; communications or knowledge-exchange owner; and data steward. Precedence: owner-confirmed adoption, non-adoption and outcome evidence outrank researcher expectation, download counts or informal praise. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: decision state, baseline and follow-up evidence must be effective-dated and attributable only to the authority that records them. Conflict rule: retain counterevidence and alternative contributors rather than assigning sole credit.

ANALYSIS REQUIRED

Operation: estimate and contribution recommendation. Decision interval: at project planning, agreed milestones and follow-up. Minimum evidence: an impact pathway, named decision or practice, baseline, dated engagement, adoption or non-adoption evidence, follow-up interval, competing influences and a prespecified outcome. Method: use contribution analysis, structured case evidence and interrupted series or matched comparison only where design supports them; compare planned and observed mechanisms. Uncertainty: report uncertainty, alternative explanations, counterevidence and the strength of contribution rather than a single attribution percentage. Safe failure: report reach or output only and state that decision or outcome change is unverified.

ANSWER BOUNDARY

Do not claim impact from dissemination alone, attribute an operational change to the study without corroboration, expose decision-makers or users, or select only favourable outcome indicators.

WHY THIS IS USEFUL

Creates credible impact evidence and helps the team invest in engagement activities that can be evaluated rather than merely counted.

RD-080

STAKEHOLDER QUESTION

CATEGORY 8

Which next study, mapping task or no-new-data option offers the greatest expected reduction in consequential uncertainty within governance limits?

SENSORS AND TELEMETRY

Proposed evidence: Use the complete evidence map from the current study: result and uncertainty summaries, failed quality gates, coverage and drift evidence, unresolved mappings, permission limits, stakeholder decisions and candidate low-risk measurements. Spatial coverage and limits: Consider only target rooms, systems and populations within current owner authority and explicitly include unobserved or underrepresented contexts. Temporal coverage and aggregation: Use the latest frozen study snapshot and current source-health and permission status; candidate pilots must specify cadence, aggregation and stopping criteria. Record-specific quality check: verify current mapping, calibration feasibility, clock integration, expected uptime, gap and drift controls and provenance plan for every proposed addition. Evidence gate: for physical inputs, carry forward verified modality and location, asset/zone mapping, commissioning, calibration or reference evidence, UTC alignment, cadence, uptime, gaps and backfill or resampling, drift and device, firmware and configuration lineage. For artefacts, retain schema, version, hash and raw-to-derived lineage; quarantine unresolved failures.

AUTHORITATIVE SUPPORTING DATA

Owners and authority: principal investigator; researchers and data scientists; data, facilities and competent professional owners; ethics, governance and security teams; stakeholders; funder and data steward. Precedence: current owner needs, permissions and unresolved evidence limits outrank novelty, sunk cost or the May 2021 plan; safety and compliance priorities remain with competent authorities. The May 2021 floor plan may support historical orientation only; current owner records and competent confirmation govern current use, access, services, hazards and closures. Time, permission and lineage: retain valid-from, observation or event, ingestion and retrieval times; record purpose and permission scope, owner, source version or hash, extraction identifier and transformations. Currency rule: needs, source health, cost and permission evidence must be current at prioritisation. Conflict rule: retain divergent stakeholder priorities and use a transparent decision rule rather than hiding disagreement.

ANALYSIS REQUIRED

Operation: recommendation using comparison and expected value of information. Decision interval: at project close and before any follow-on commitment. Minimum evidence: a finite option set, remaining decision uncertainty, stakeholder value, scientific novelty, feasibility, cost, privacy and security risk, transfer need, expected information gain and explicit stop conditions. Method: score options with sensitivity analysis, dominance checks and expected value of information; include no-new-study and metadata-only options and test whether ranking changes under plausible weights. Uncertainty: report ranking stability, ranges, unresolved dependencies and the evidence needed before the top option can start. Safe failure: recommend no further sensing, a narrow pilot, a mapping or metadata task, external replication or no follow-on study.

ANSWER BOUNDARY

Do not prioritise novelty over decision value, promise deployment or validation, bypass owners and approvals, conceal uncertainty or continue collection merely because infrastructure exists.

WHY THIS IS USEFUL

Turns the full catalogue into an accountable next-step choice: a focused study, a mapping or metadata task, external replication, a lower-risk pilot or no further collection.

Catalogue summary: from a research question to a defensible result

The 80 questions form one evidence chain. A defensible answer requires an explicit study boundary, verified measurement, authorised supporting data, a method matched to dependence and uncertainty, and communication that stays within professional and ethical authority.

1

RESEARCH QUESTIONS, HYPOTHESES AND STUDY BOUNDARIES
Define a testable estimand, comparison, precision target and transfer boundary.

2

DATA CATALOGUE, VARIABLES, COVERAGE AND ACCESS
Establish which evidence exists, what variables mean, where they cover and who may use them.

3

SENSOR PLACEMENT, CALIBRATION AND DATA QUALITY
Demonstrate representativeness, calibration, timing integrity, continuity and comparability.

4

LINKAGE, TIMESTAMPS, PROVENANCE AND VERSIONING
Create trustworthy joins across identifiers, time semantics, spatial states and immutable lineage.

5

METHODS, BASELINES, CONFOUNDING, BIAS AND UNCERTAINTY
Choose methods and baselines that address dependence, confounding, bias and decision uncertainty.

6

PRIVACY, ETHICS, CONSENT AND DISCLOSURE CONTROL
Minimise data, respect approvals and consent, and prevent disclosure or sensitive inference.

7

REPRODUCIBILITY, CODE, METADATA AND SECURE SHARING
Make extraction, code, environments, tests, sharing and preservation reproducible and secure.

8

INTERPRETATION, VISUALISATION, PUBLICATION AND IMPACT
Communicate uncertainty, align claims with evidence and plan responsible publication and impact.

DEFAULT SAFE ANSWER SEQUENCE

ASK

VERIFY

AUTHORISE

ANALYSE

QUANTIFY

BOUND

ACT THROUGH OWNER

Evidence, permission and safe-answer rules

Use these rules with every question card. A precise refusal or a narrower descriptive answer is a valid research result when the required evidence or authority is missing.

EVIDENCE GATES

- Call a signal observed only when deployment, asset mapping, commissioning and current health are independently confirmed.
- Preserve native values, units, calibration, clock quality, uptime, gaps, drift and immutable provenance before aggregation.
- Distinguish lookup, observation, calculation, estimate, forecast, comparison, candidate diagnosis and recommendation in every answer.

AUTHORITATIVE PRECEDENCE

- Current source owners and competent professionals decide room use, assets, services, access, hazards, closures, safety, compliance and action.
- The May 2021 floor plan is historical spatial reference only and cannot override current records or competent confirmation.
- Keep effective, observed, ingested and retrieved times; quarantine unresolved conflicts rather than selecting a convenient source.

PERMISSION AND PRIVACY

- Use least privilege, purpose-limited access, minimum resolution and approved retention; technical reachability is not authorisation.
- Do not identify or track people, infer sensitive traits, inspect speech or network content, or reconstruct individual routines.
- When disclosure risk cannot be reduced proportionately, aggregate, restrict access, use safe fixtures or do not release.

ANALYSIS AND COMMUNICATION

- Require minimum independent evidence, method-appropriate uncertainty, sensitivity to missingness and a declared safe failure.
- Do not fabricate missing data, conceal uncertainty, present correlation as causation or claim unverified validation.
- Research may inform an owner decision; it cannot certify, authorise, promise or directly control operational, safety or compliance action.

ASK PRECISELY | VERIFY PHYSICALLY | AUTHORISE EXPLICITLY | QUANTIFY UNCERTAINTY | STOP SAFELY

Talking Abacws supports evidence-bounded research. Current owners and competent professionals always retain operational authority.